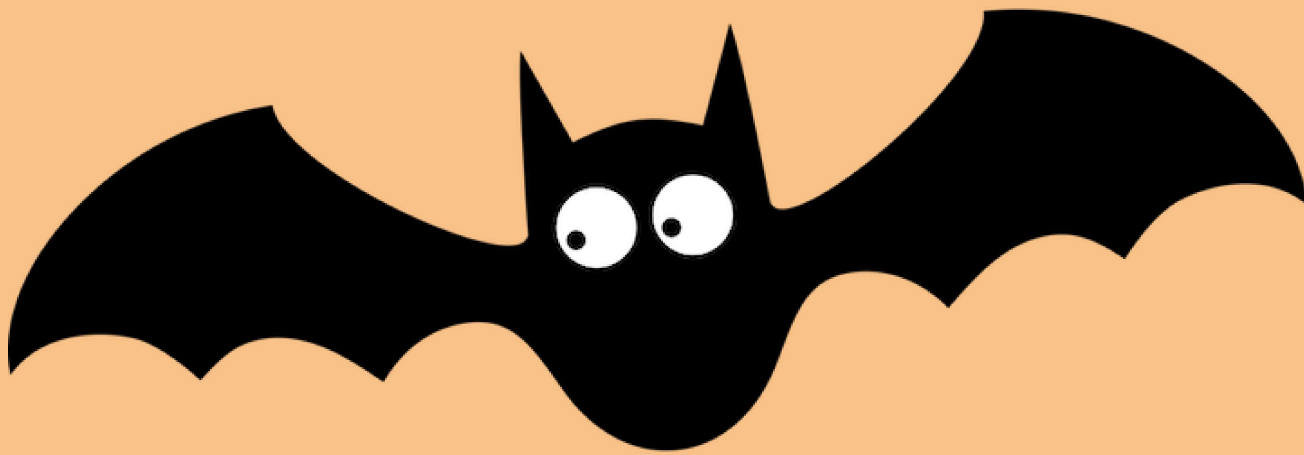




Fake Data

Son Nguyen
September 28, 2020

Creating Useful

Classification Problem



or



Classification Problem



Classification Problem

- Credit Card Default?



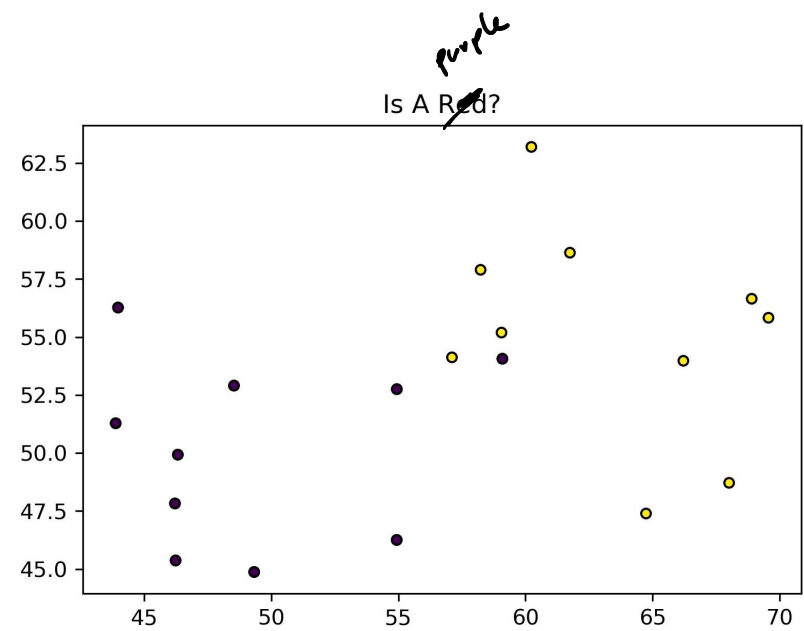
Classification Problem

- Credit Card Default?
- Or not?



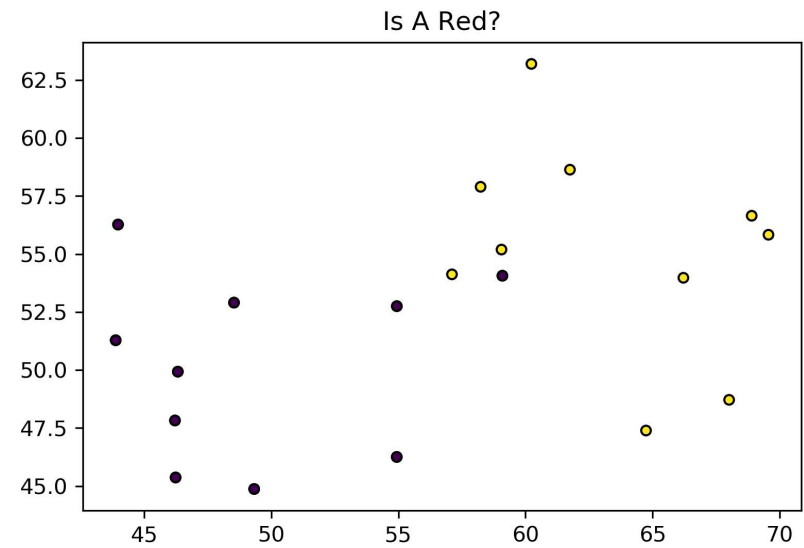
Classification Problem

- Is A yellow?



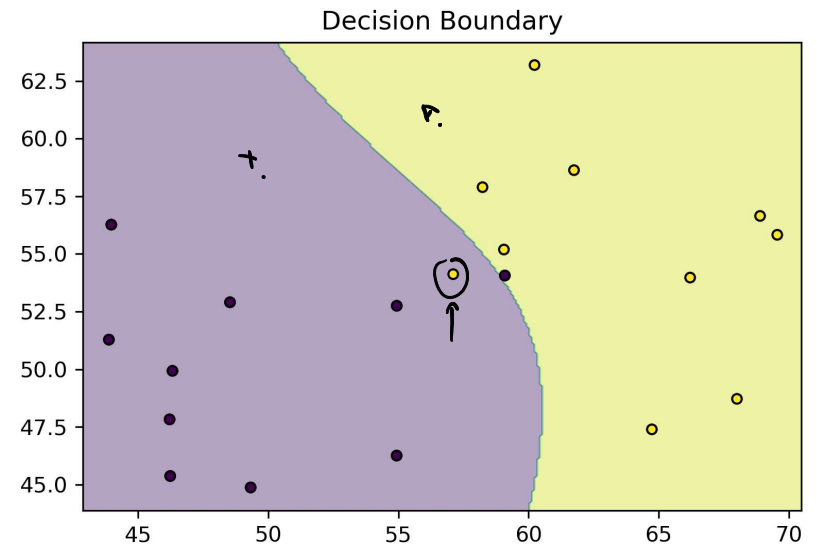
Classification Problem

- Is A yellow?
- Or purple?



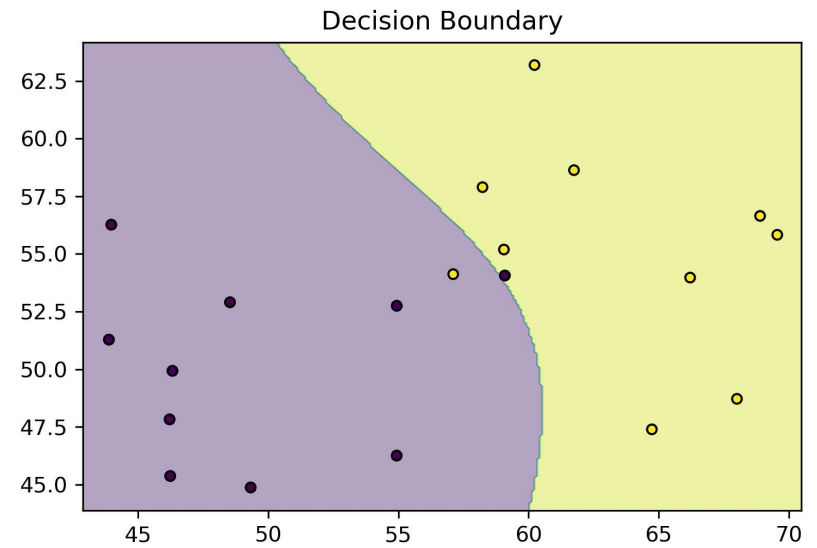
Classification Problem

- Decision Boundary



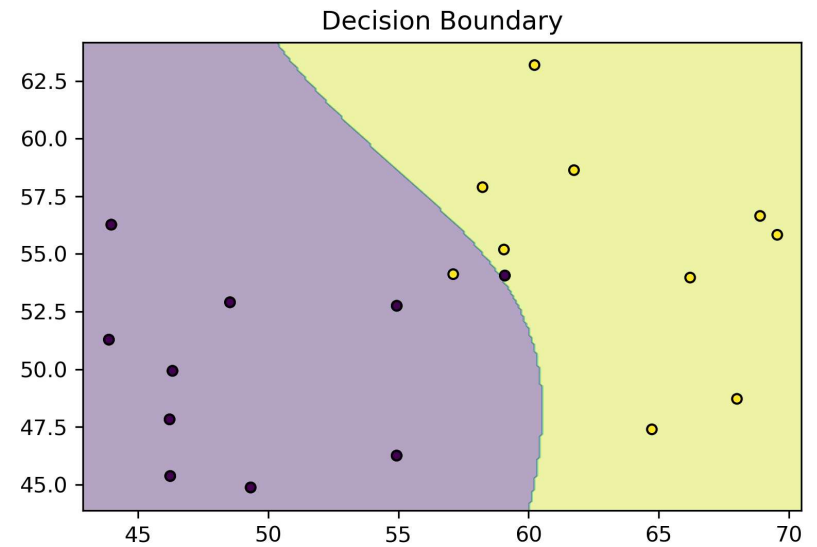
Classification Problem

- Decision Boundary
- of a classification model



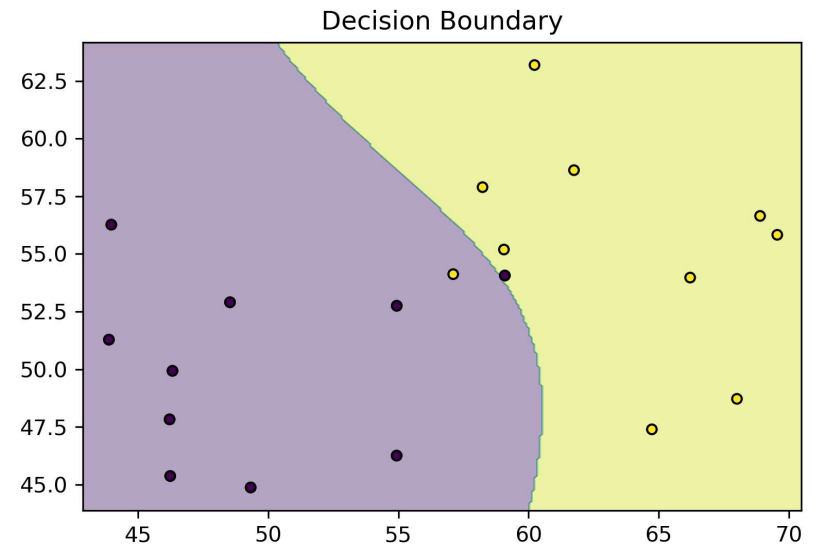
Classification Problem

- Decision Boundary
- of a classification model
- A is...

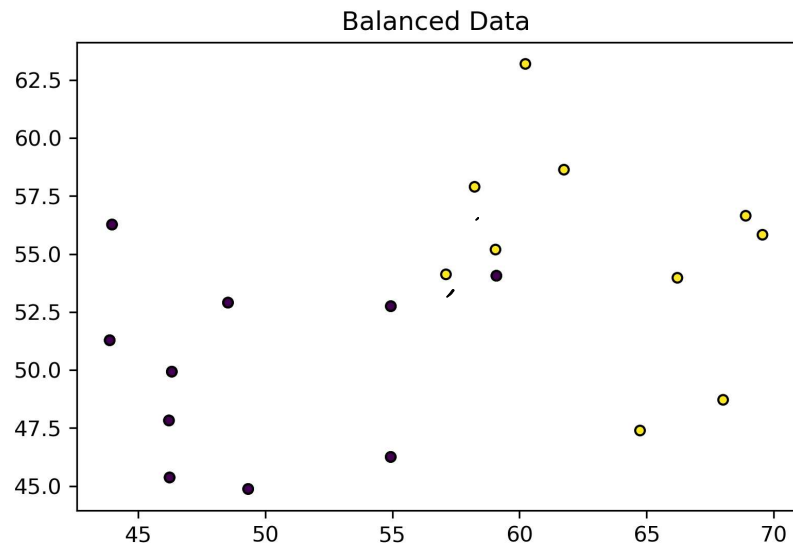


Accuracy

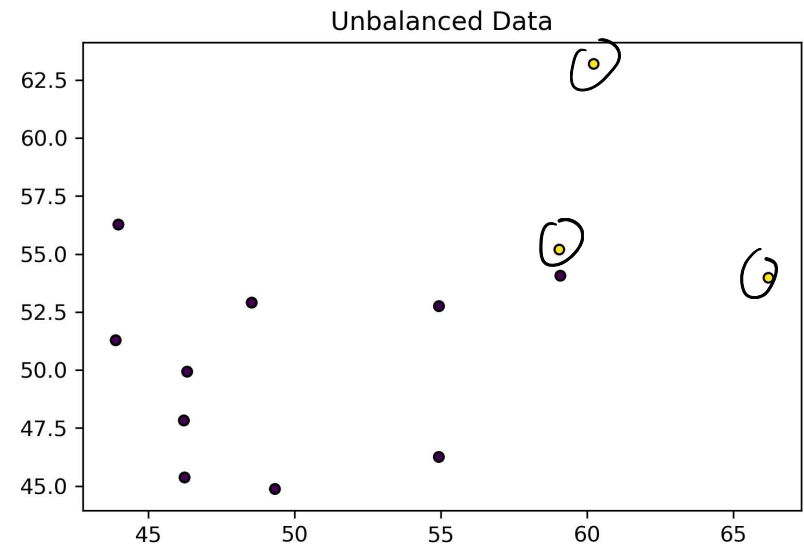
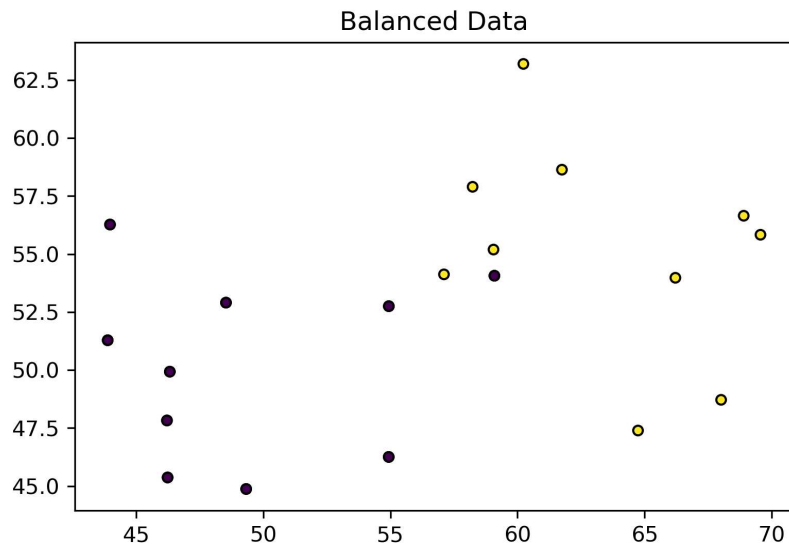
Purple Accuracy	Yellow Accuracy	Accuracy	Balanced Accuracy
0.9	0.9	0.9	0.9



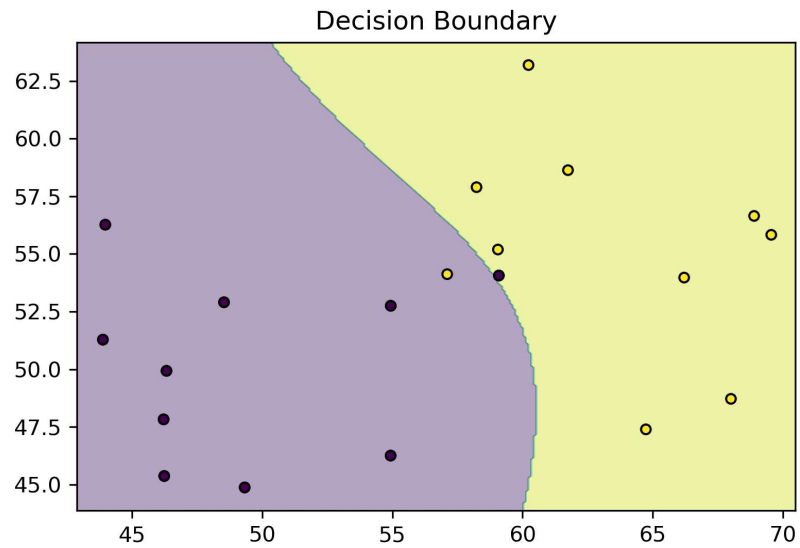
Balance vs. Unbalanced Data



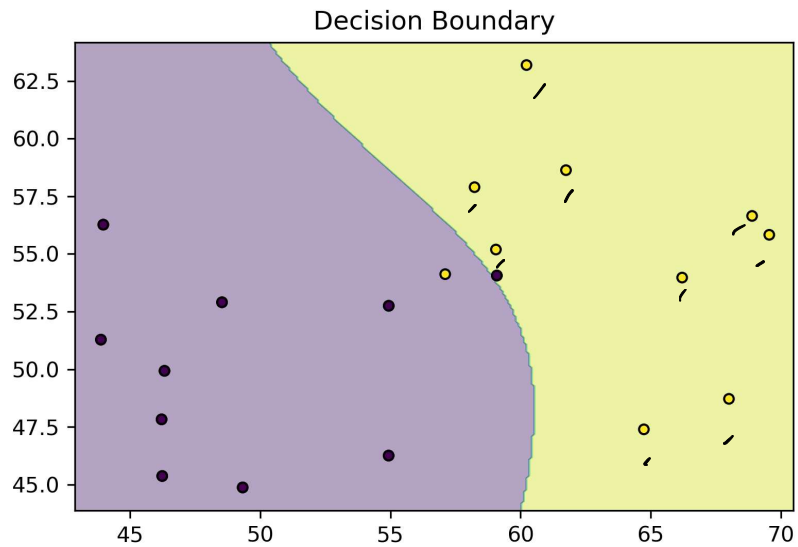
Balance vs. Unbalanced Data



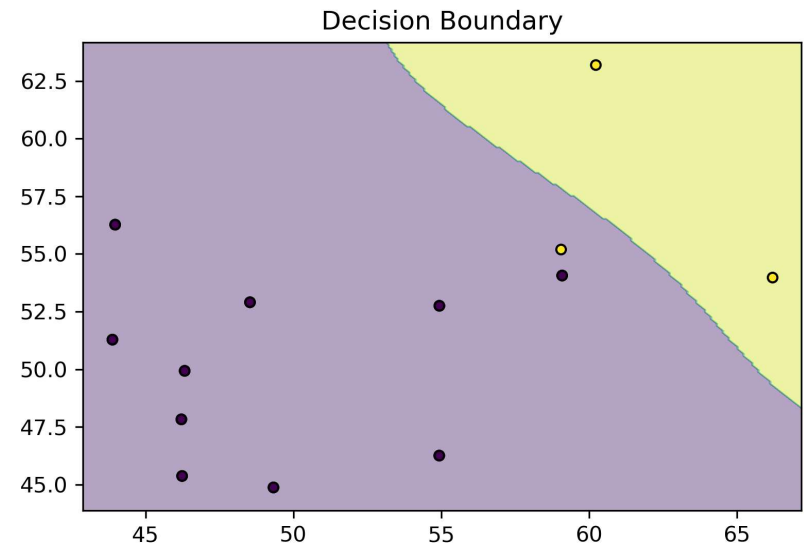
Decision Boundary of Unbalanced Data



Decision Boundary of Unbalanced Data

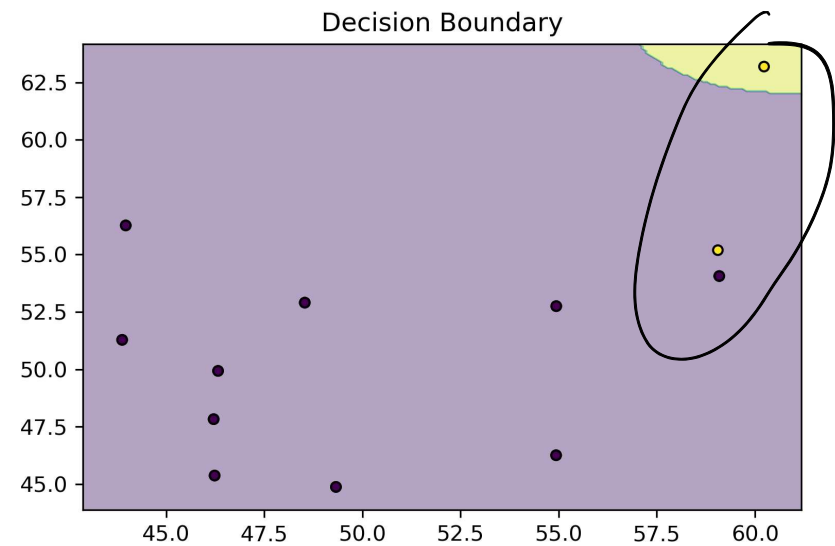
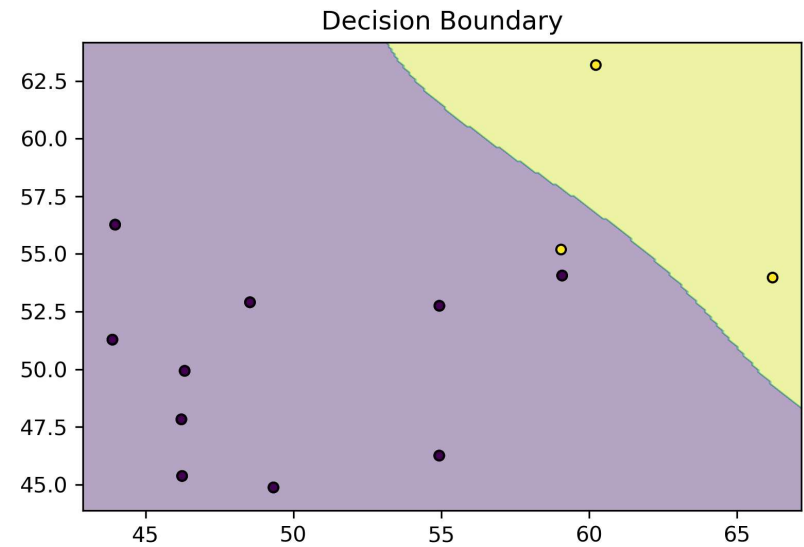
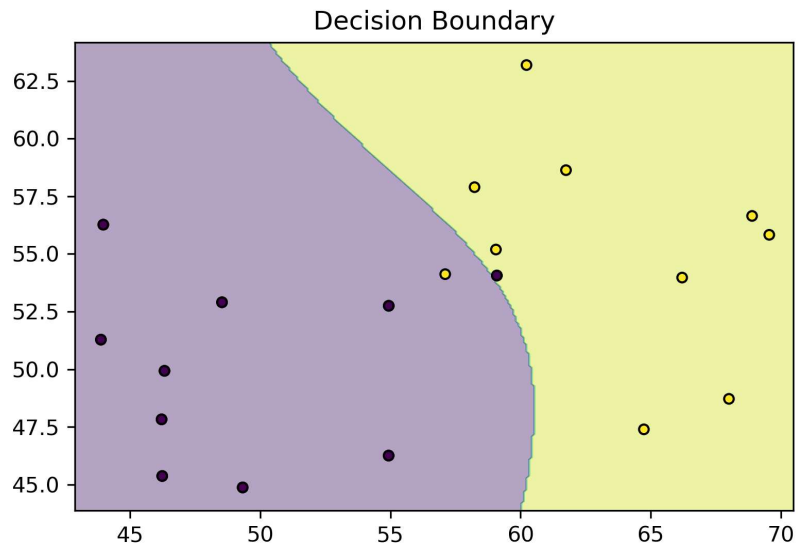


9/10
for yellow
90%

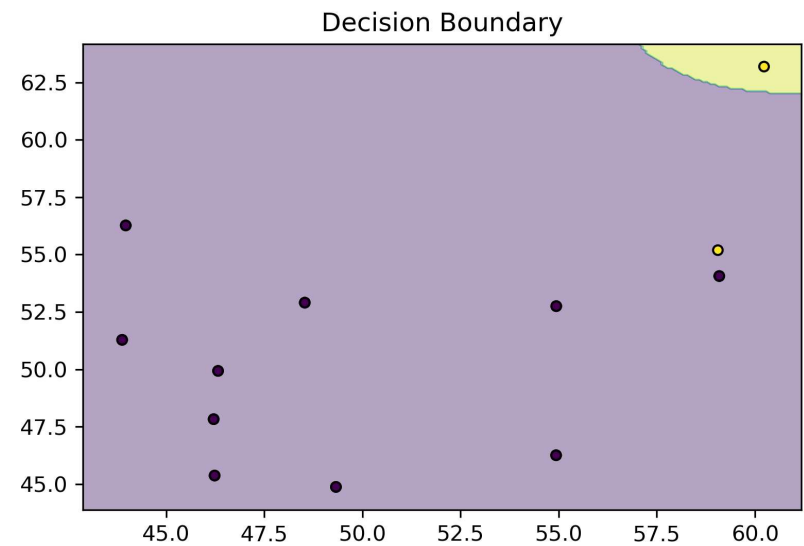
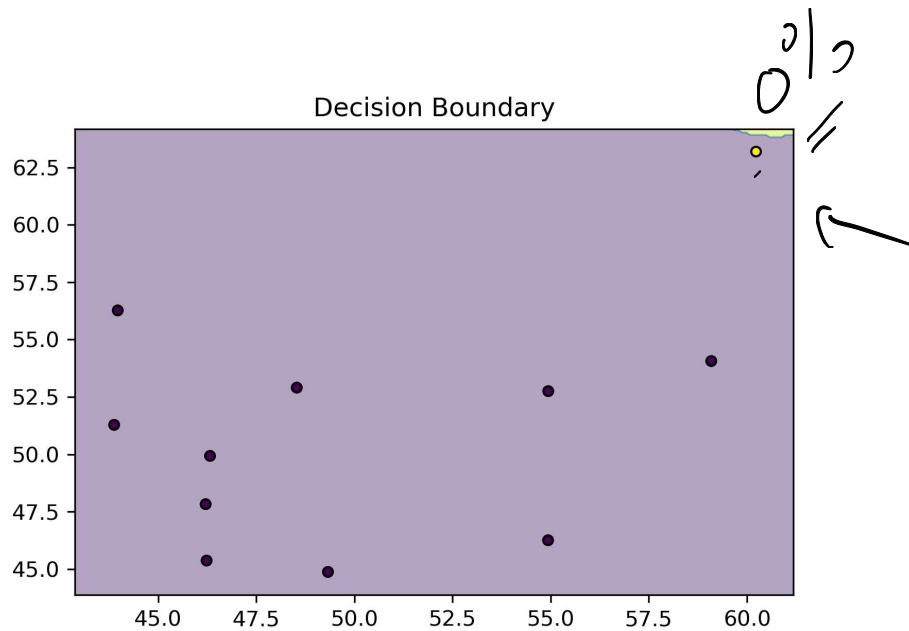
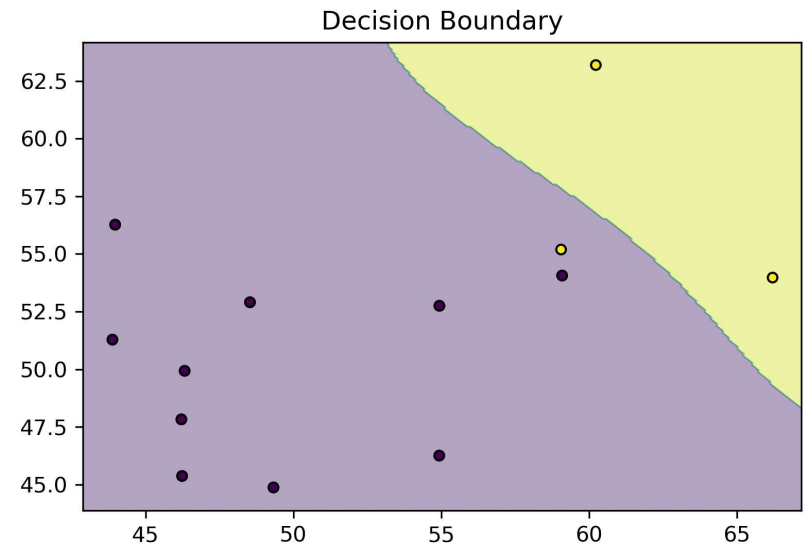
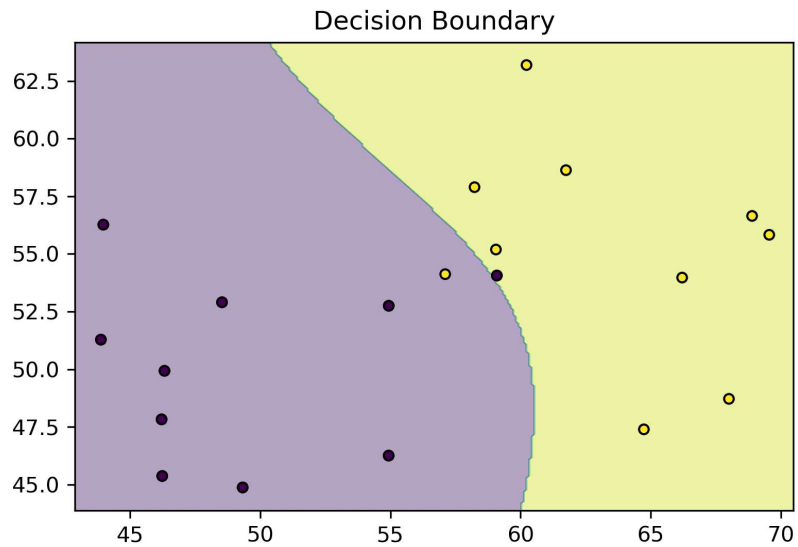


2/3 = 66.6%

Decision Boundary of Unbalanced Data

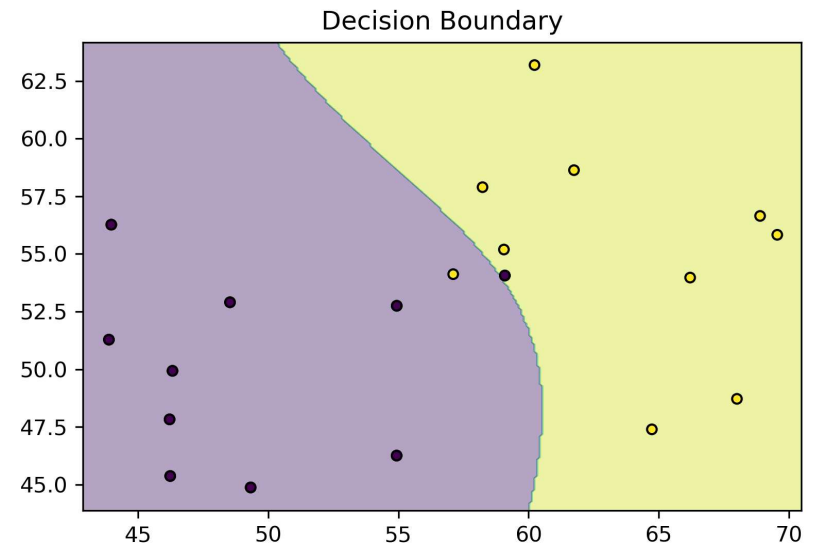


Decision Boundary of Unbalanced Data



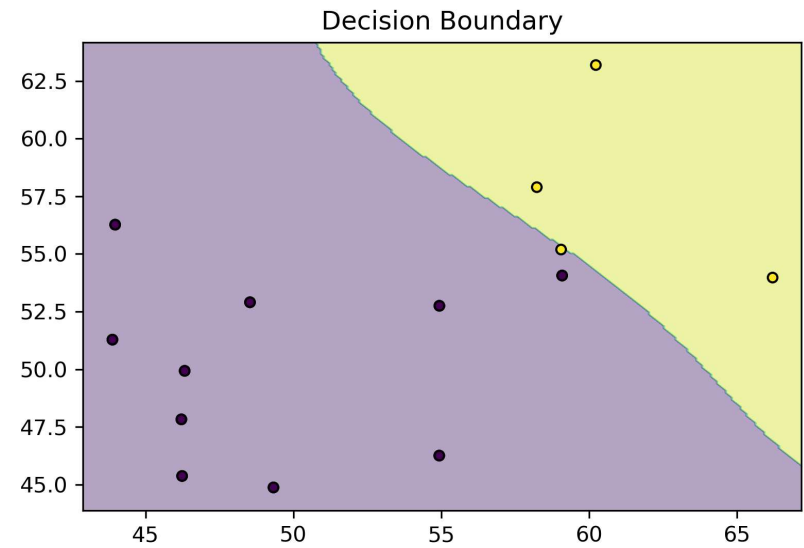
Accuracy

Yellow Accuracy	Purple Accuracy	Accuracy	Balanced Accuracy
0.9	0.9	0.9	0.9



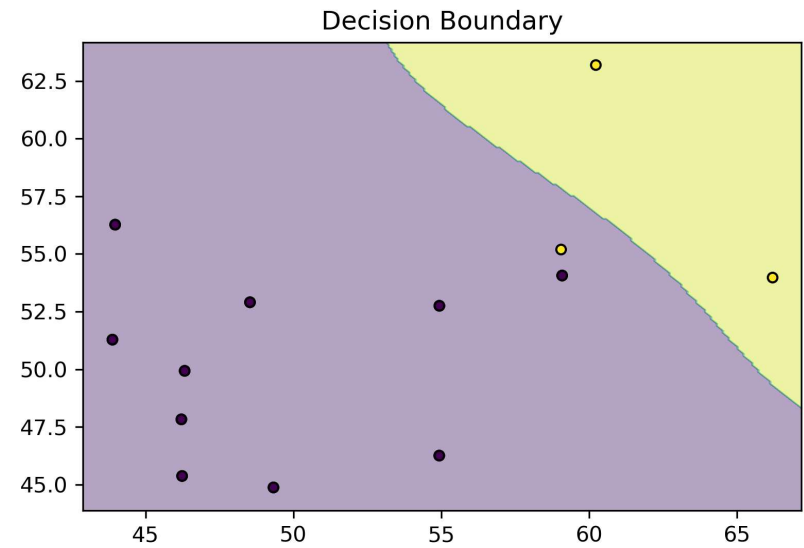
Accuracy

Yellow Accuracy	Purple Accuracy	Accuracy	Balanced Accuracy
0.75	1	0.958333	0.875



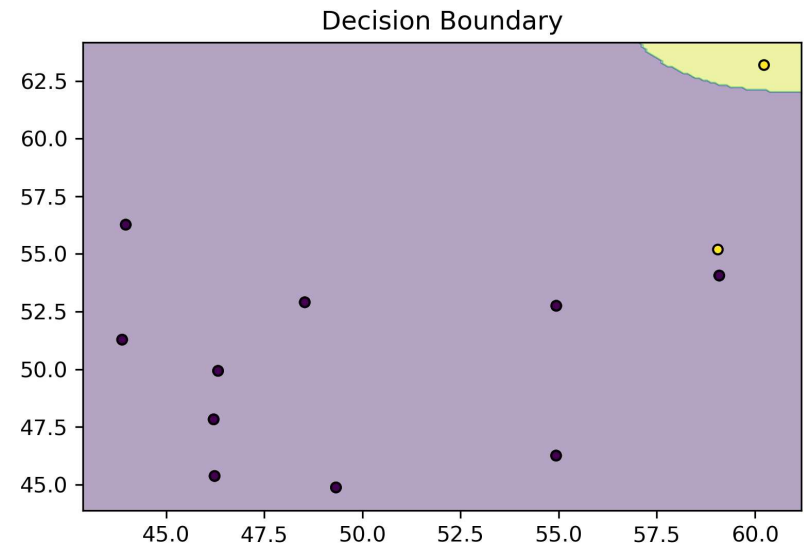
Accuracy

Yellow Accuracy	Purple Accuracy	Accuracy	Balanced Accuracy
0.666667	1	0.956522	0.833333



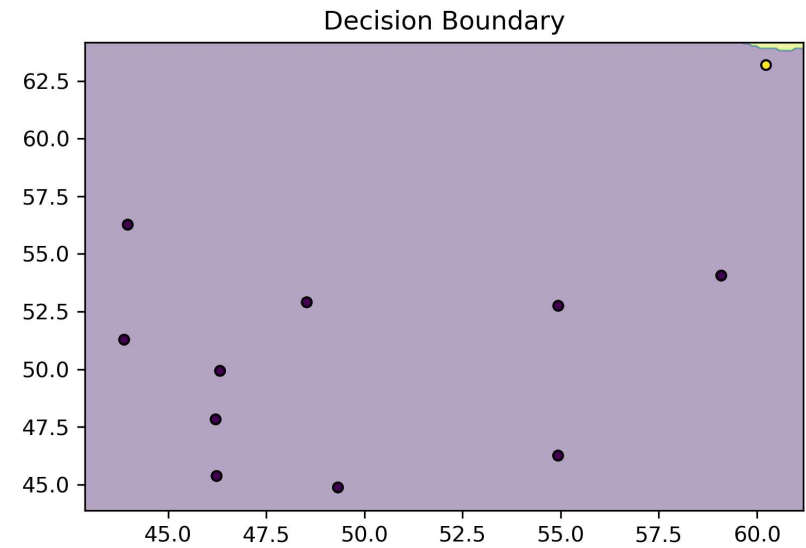
Accuracy

Yellow Accuracy	Purple Accuracy	Accuracy	Balanced Accuracy
0.5	1	0.954545	0.75



Accuracy

Yellow Accuracy	Purple Accuracy	Accuracy	Balanced Accuracy
0	1	0.952381	0.5



- It's more difficult to detect yellow points when the imbalance increases
- Yellow points could present
 - Cancer Patients,
 - Credit Card Fraud,
 - Bankrupted Companies,
 - Terrorist
- They are all rare events.

Question

- How to detect the yellow points when there are too few of them?

Strategy 1

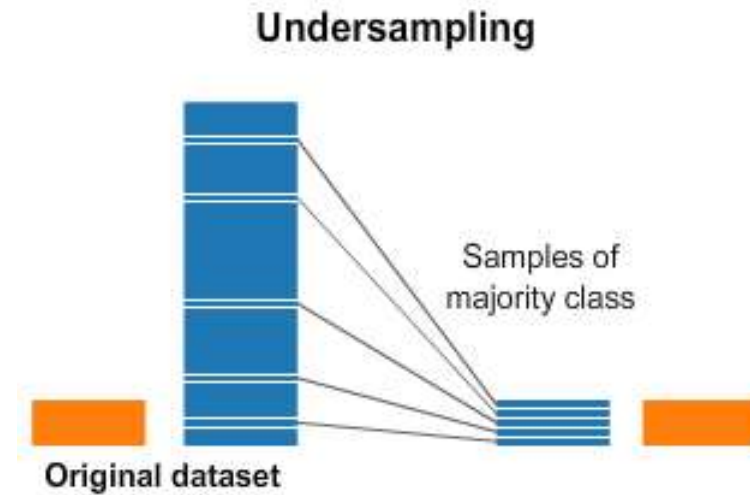
- Remove the purple points so that the data more balanced

Strategy 1

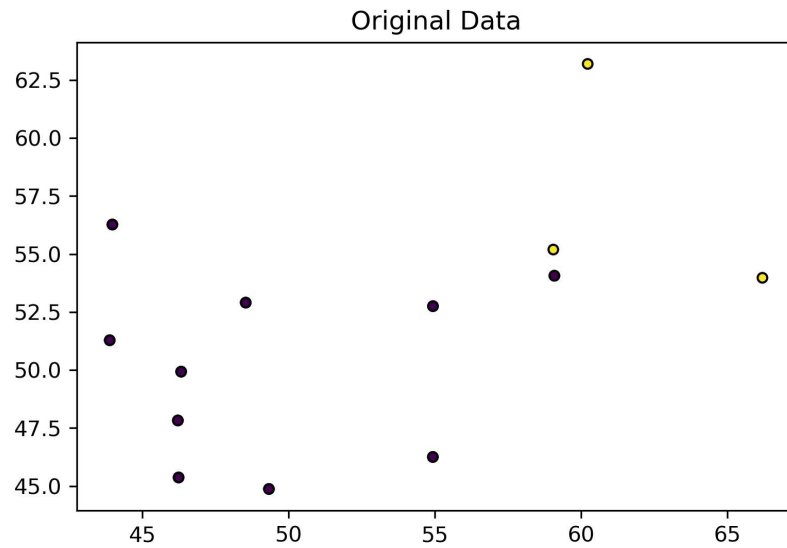
- Remove the purple points so that the data more balanced
- This is called under-sampling

Strategy 1

- Remove the purple points so that the data more balanced
- This is called under-sampling

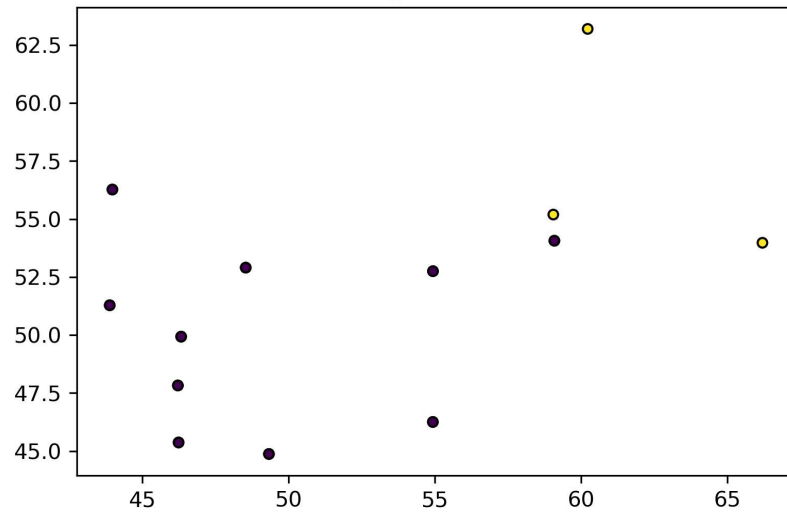


Undersampling

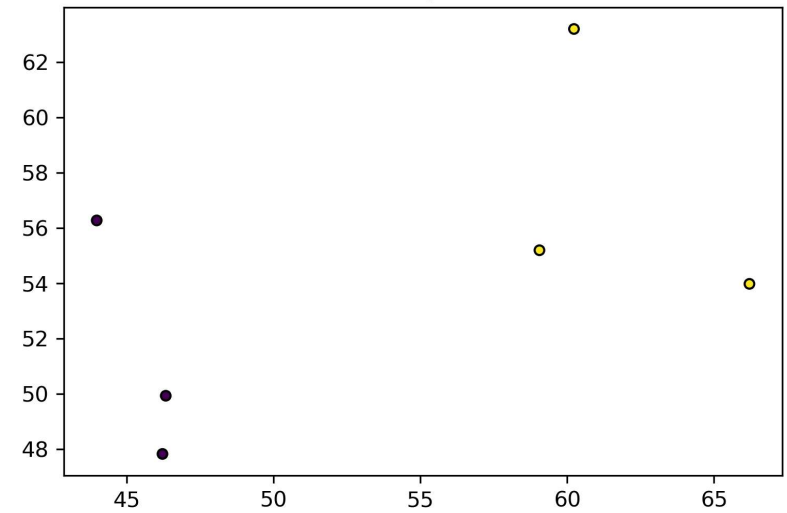


Undersampling

Original Data

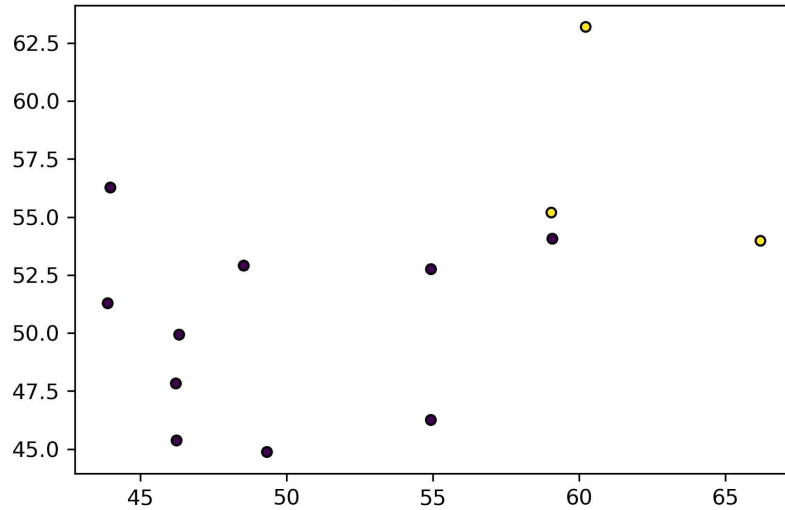


Undersampled Data

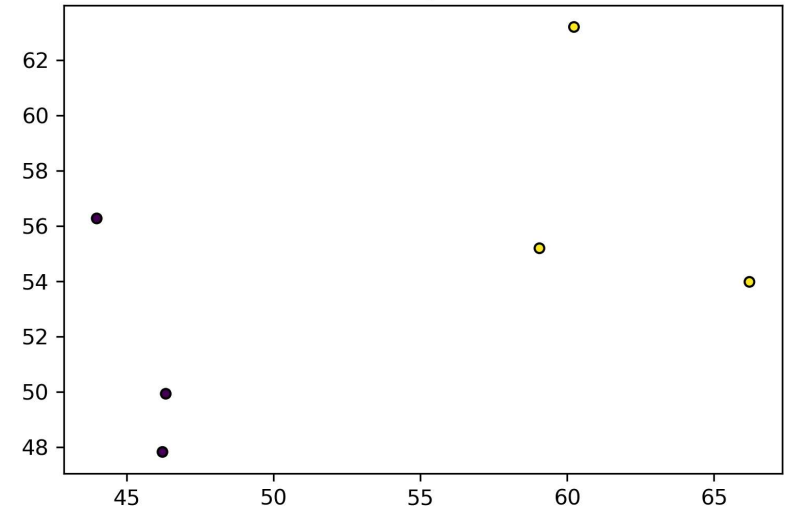


Undersampling

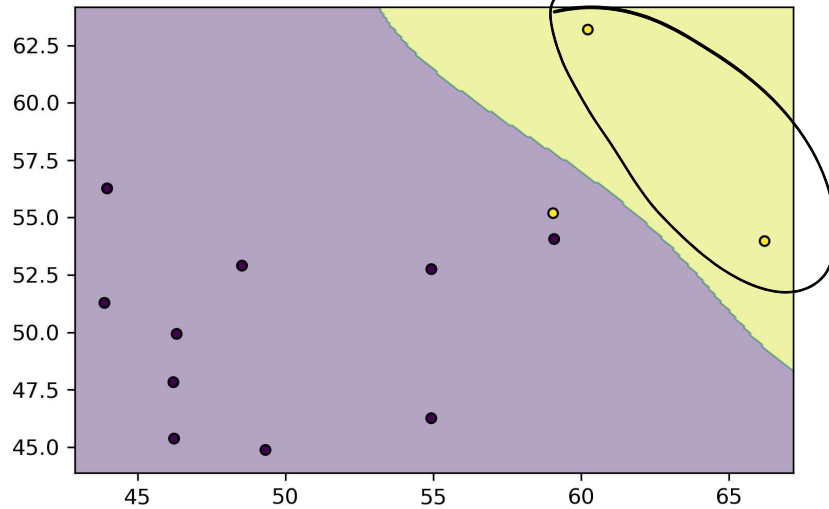
Original Data



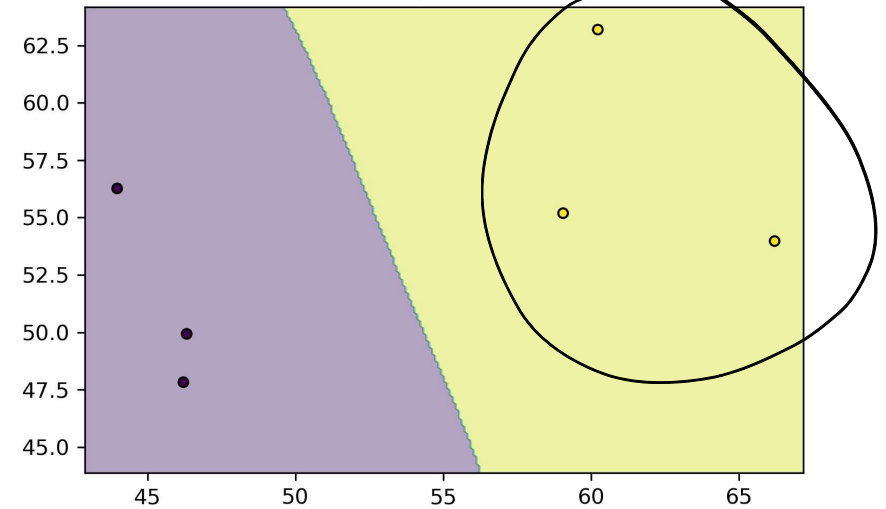
Undersampled Data



Original Boundary

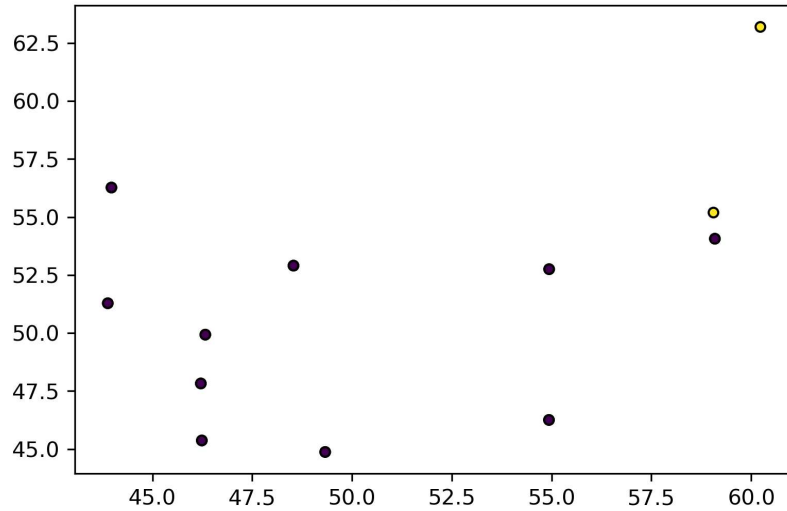


Undersampled Boundary

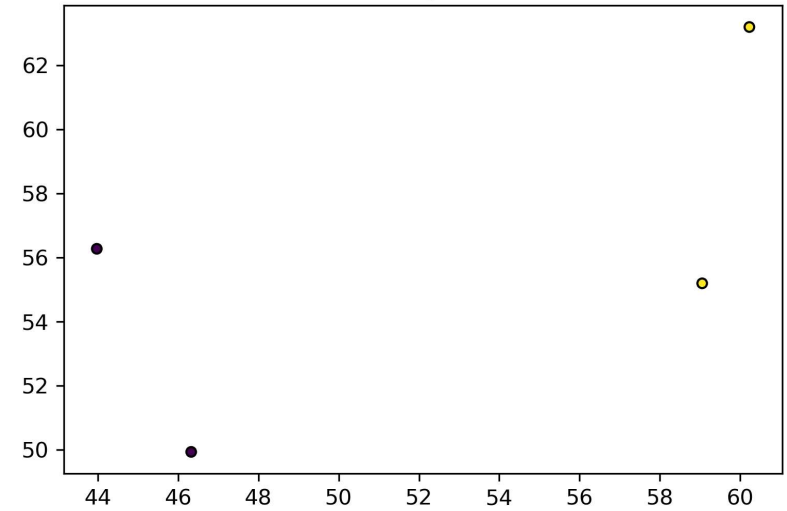


Undersampling

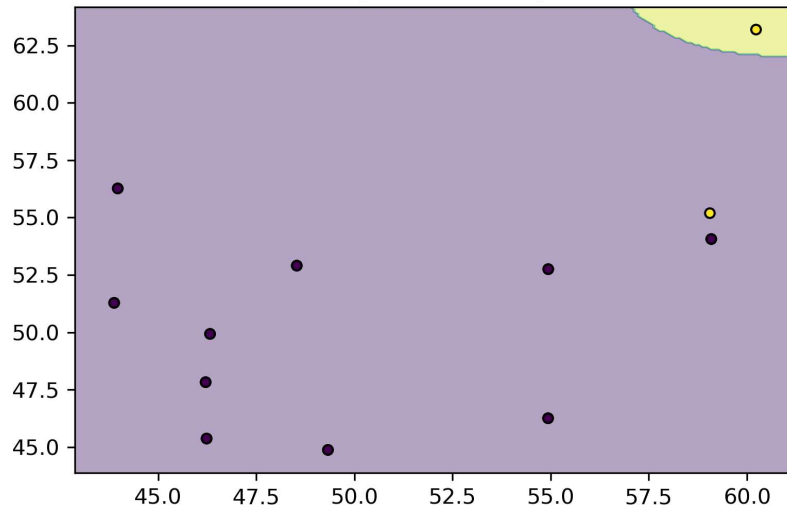
Original Data



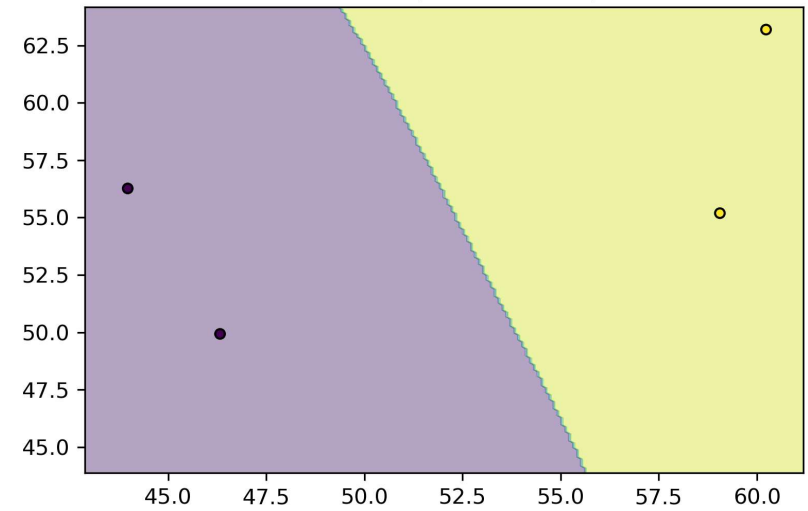
Undersampled Data



Original Boundary

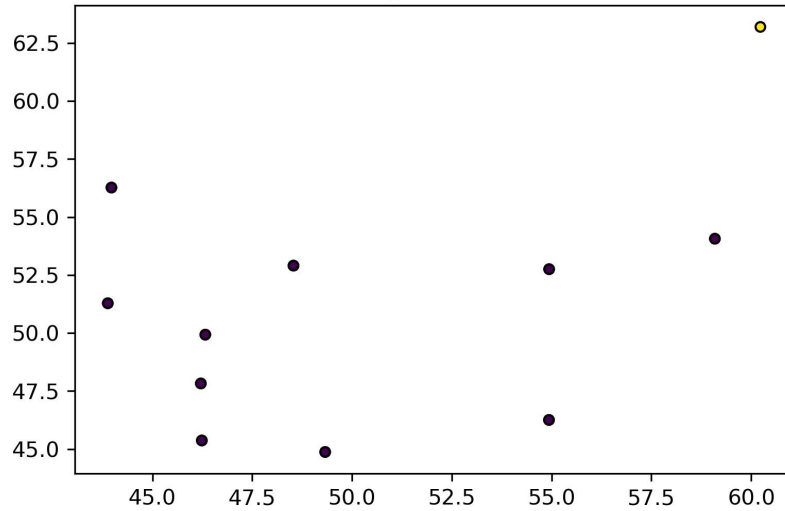


Undersampled Boundary

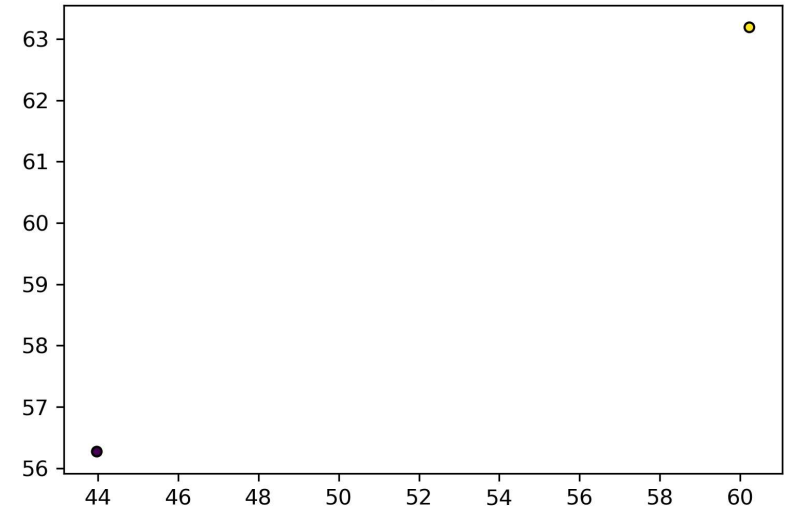


Undersampling

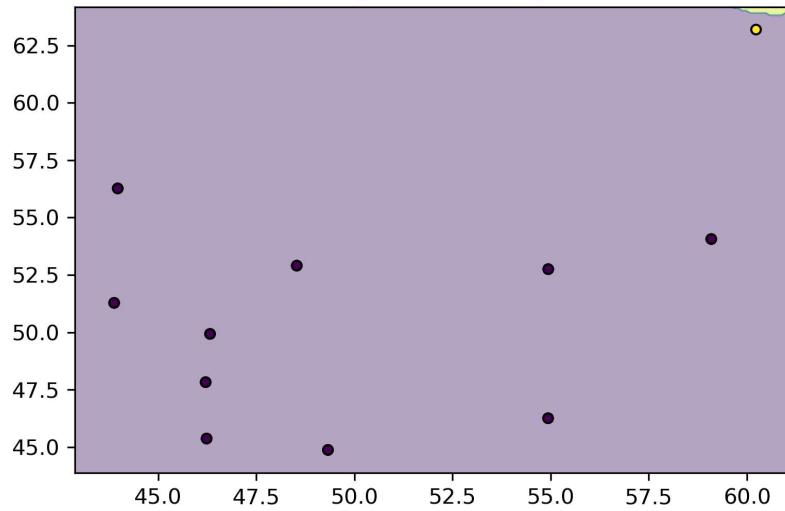
Original Data



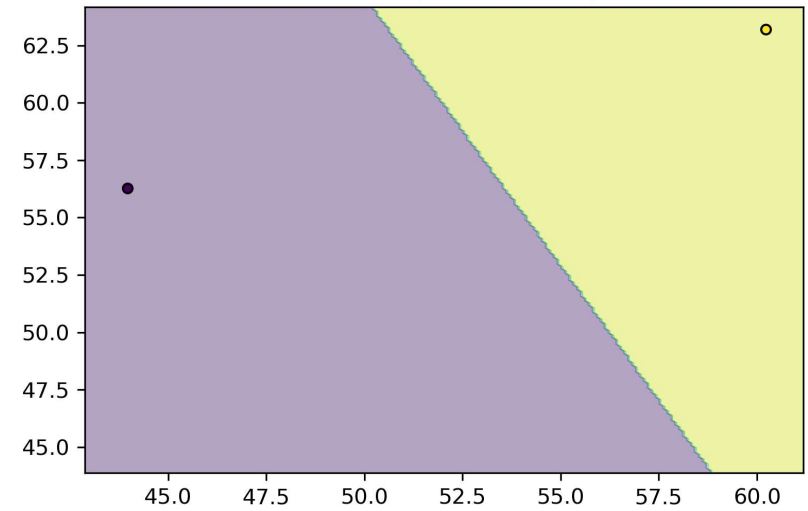
Undersampled Data



Original Boundary

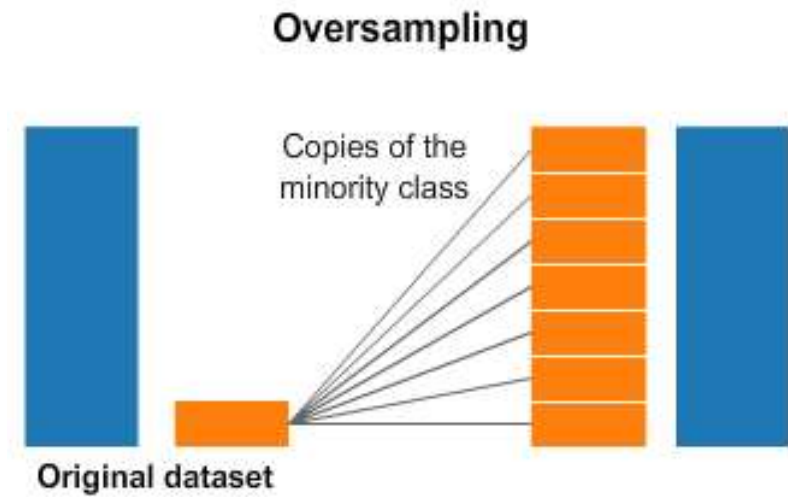


Undersampled Boundary

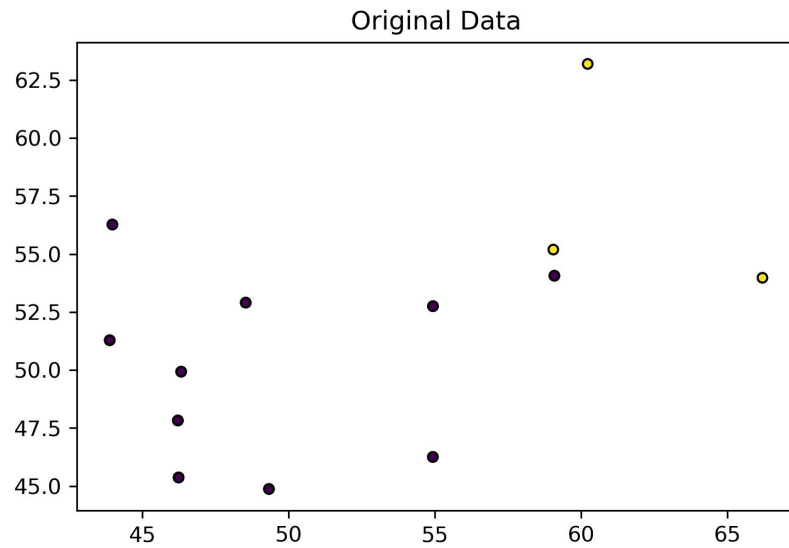


Strategy 2

- Duplicate the yellow points many times so that the data more balanced
- This is called Over-sampling

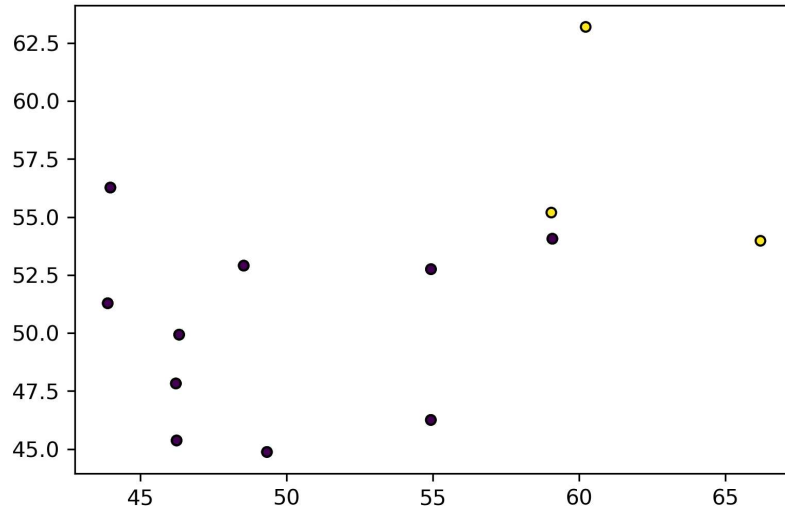


Oversampling

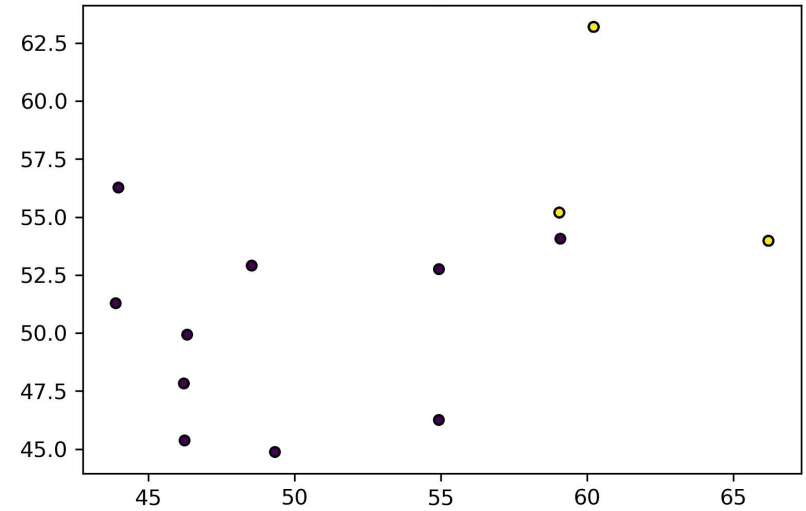


Oversampling

Original Data

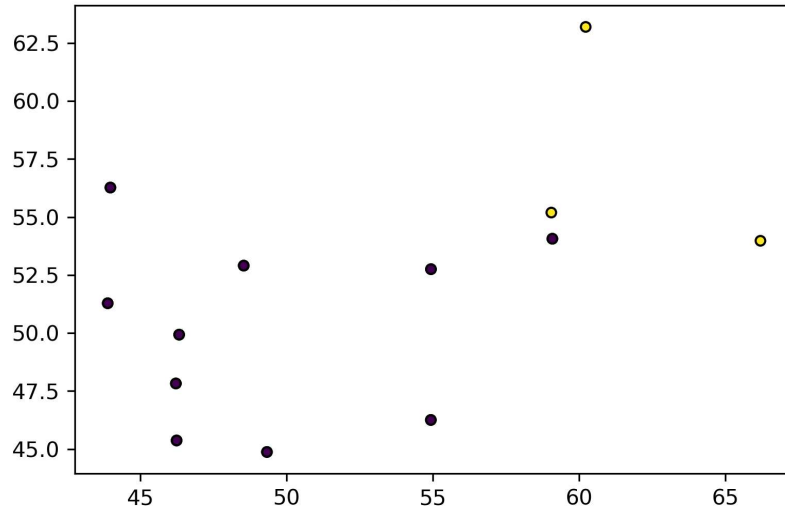


Oversampled Data

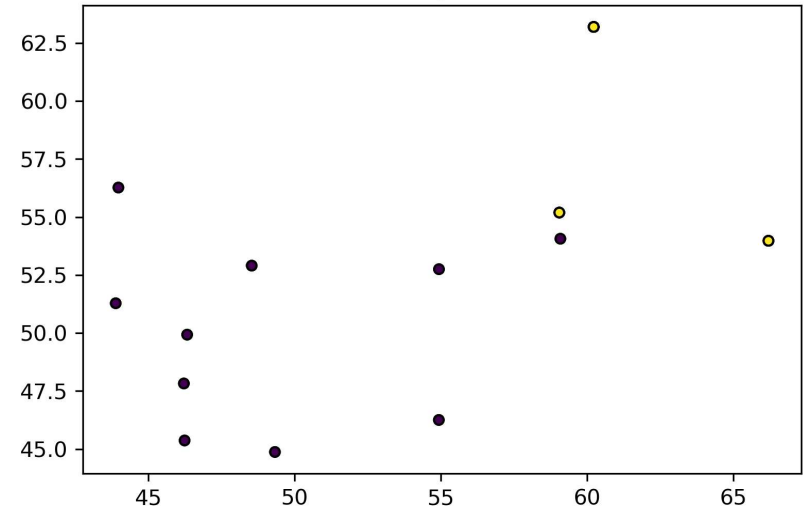


Oversampling

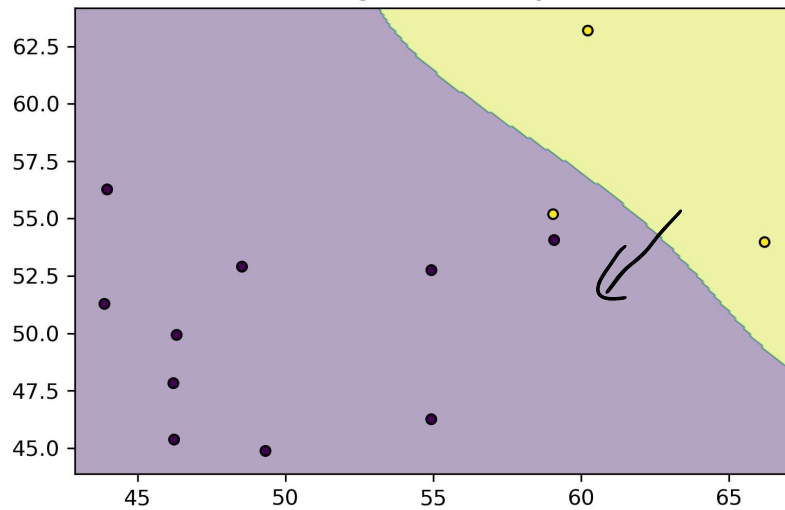
Original Data



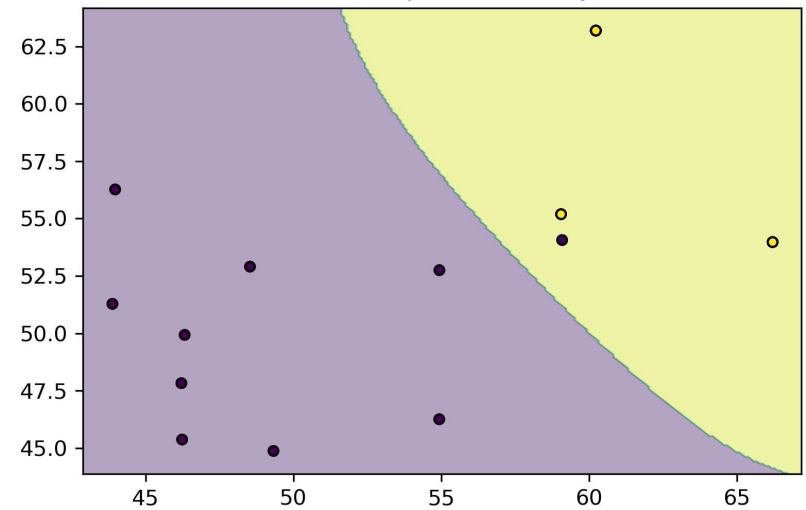
Oversampled Data



Original Boundary

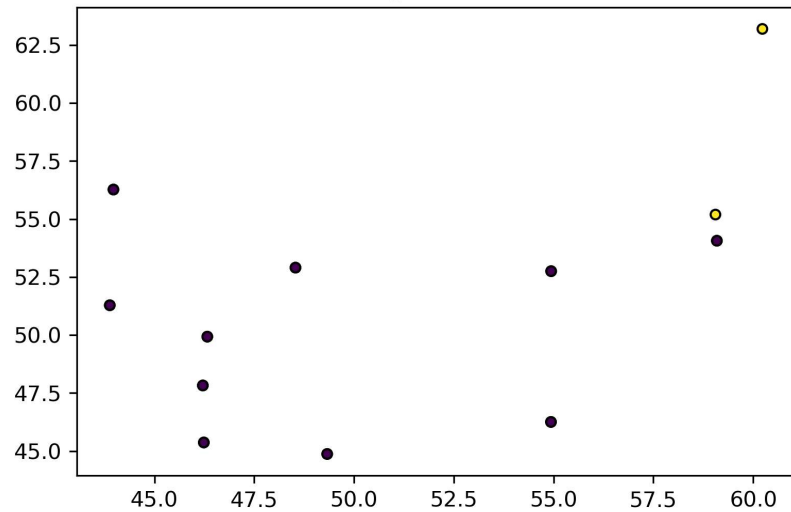


Oversampled Boundary

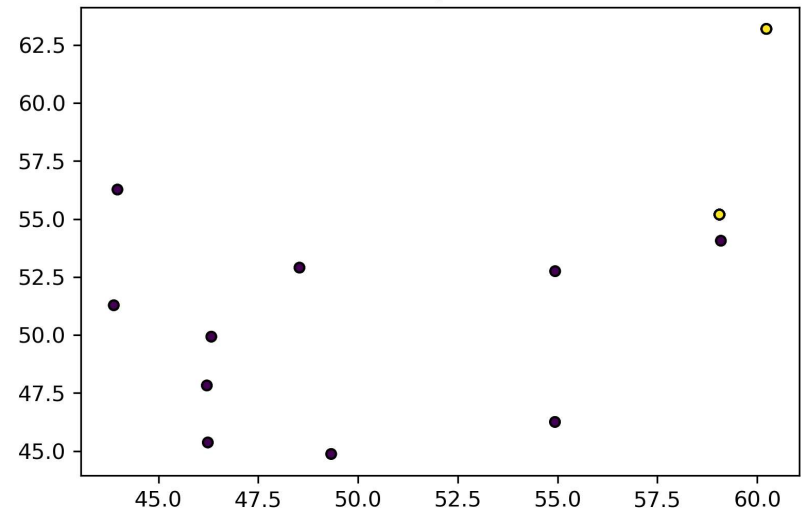


Oversampling

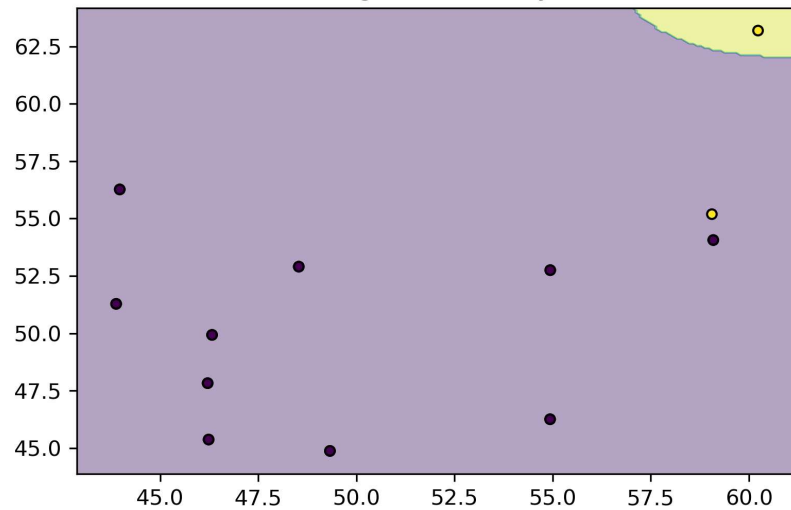
Original Data



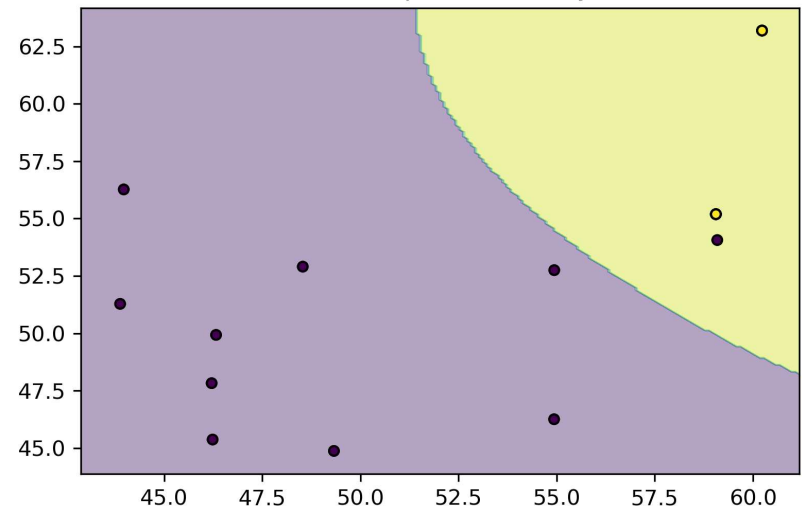
Oversampled Data



Original Boundary

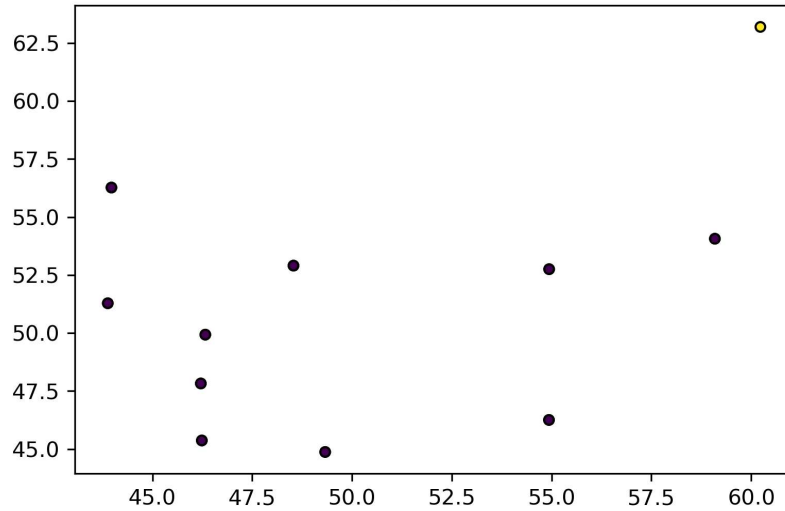


Oversampled Boundary

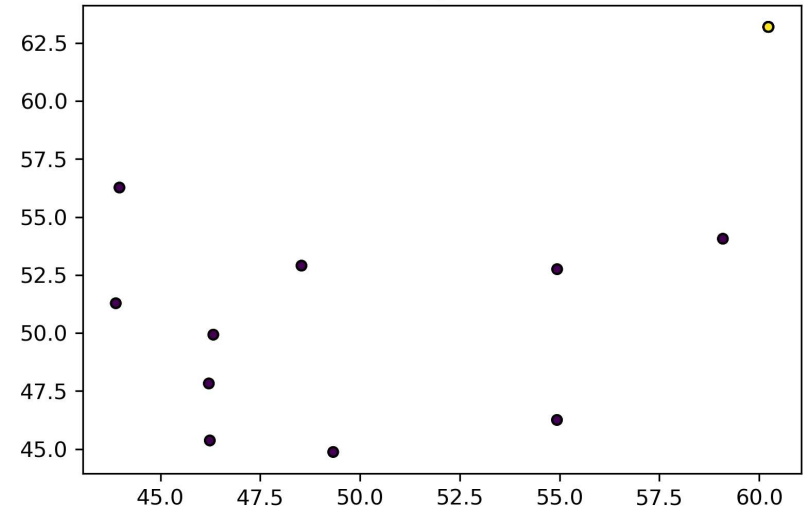


Oversampling

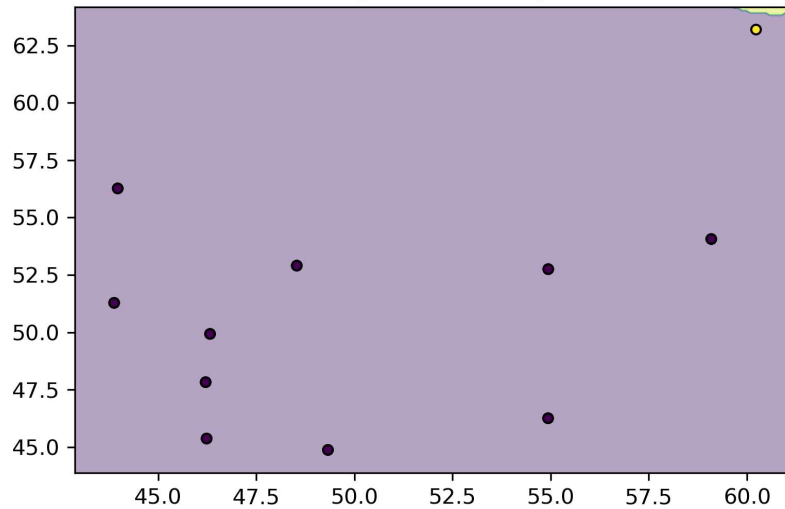
Original Data



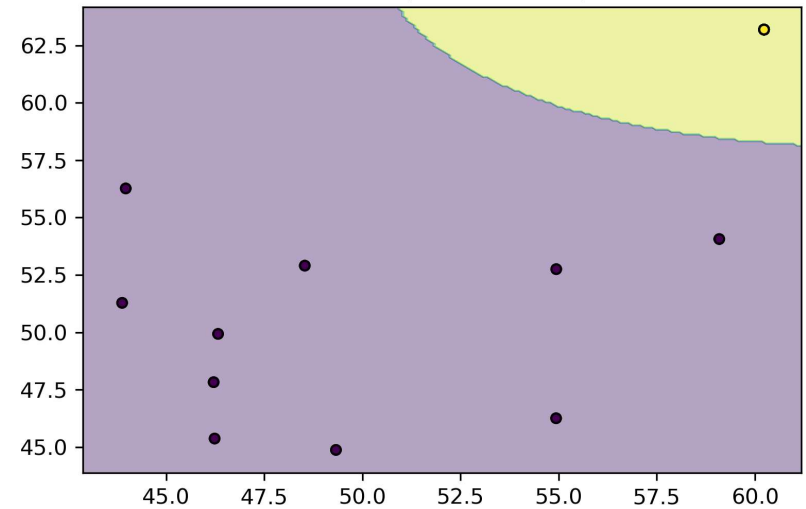
Oversampled Data



Original Boundary



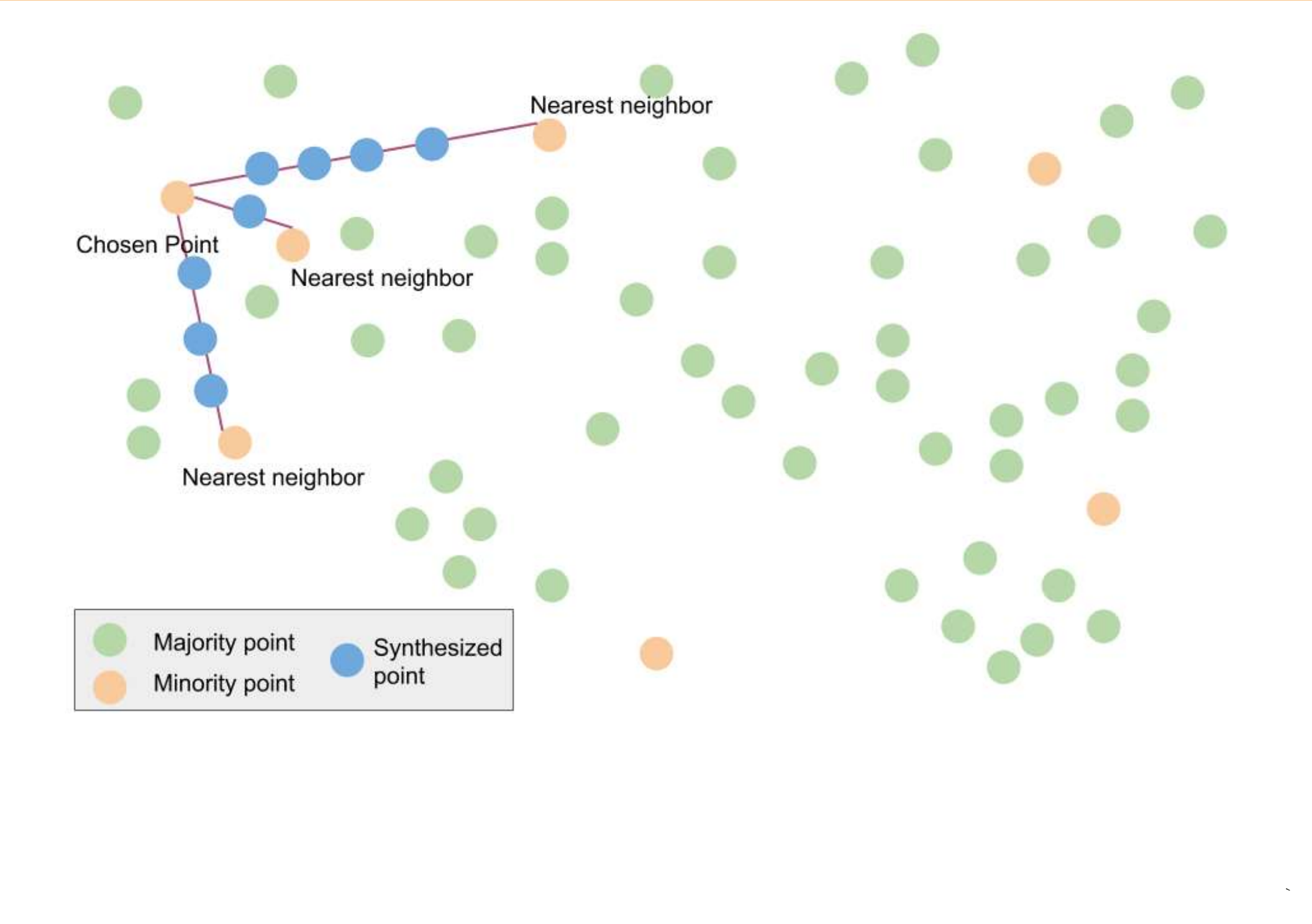
Oversampled Data



Strategy 3:

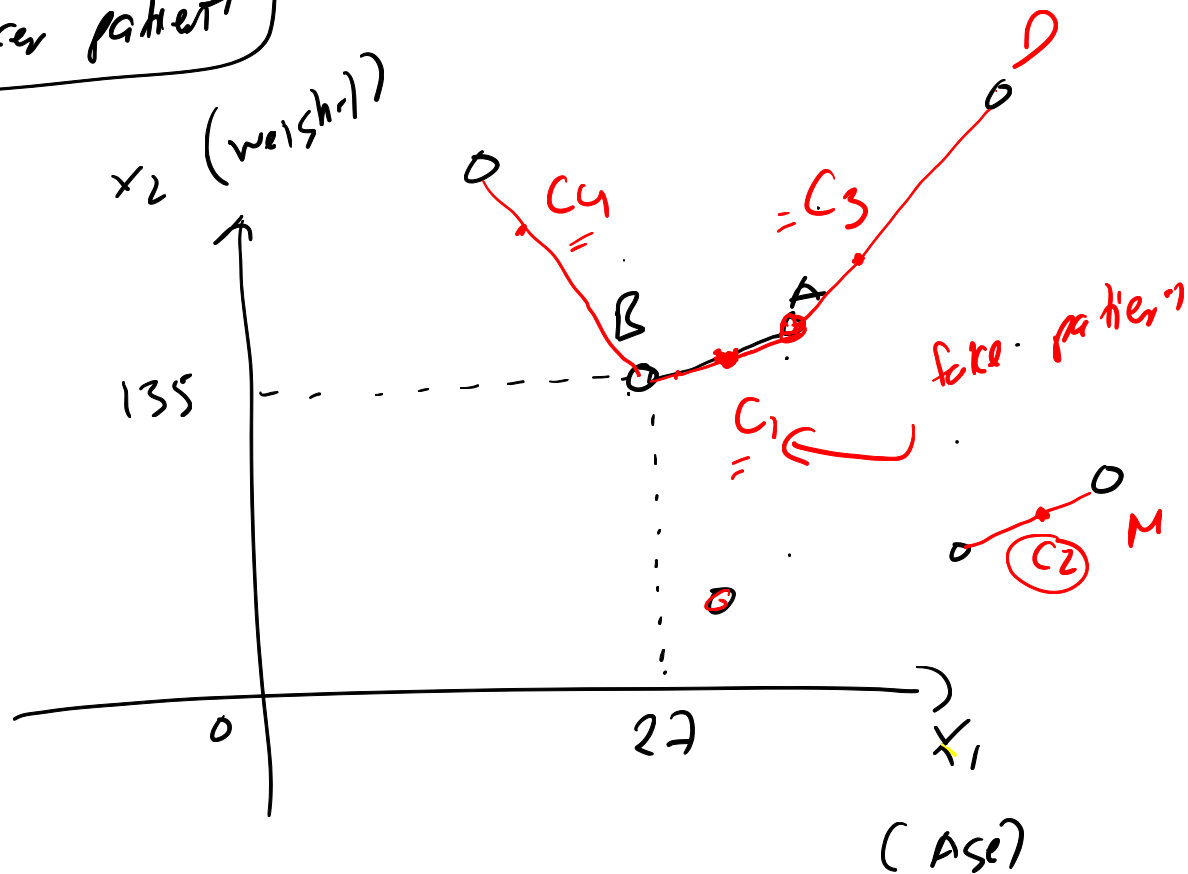
- Generate fake ~~purple~~ points

yellow



1 yellow point = 1 cancer patient

purple = not cancer



SMOTE

(i) pick 1 closest to A (nearest n.)

Input

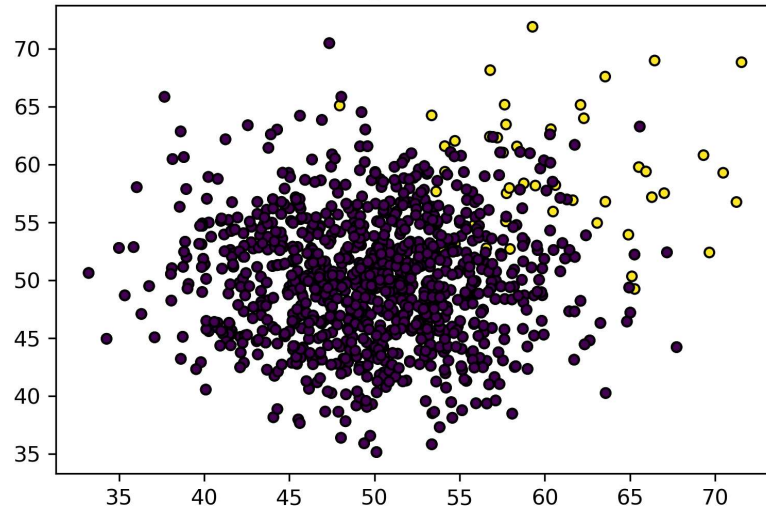
	x_1	x_2
1	2	3
3	6	7
2	$\vdots$	$\vdots$

paths

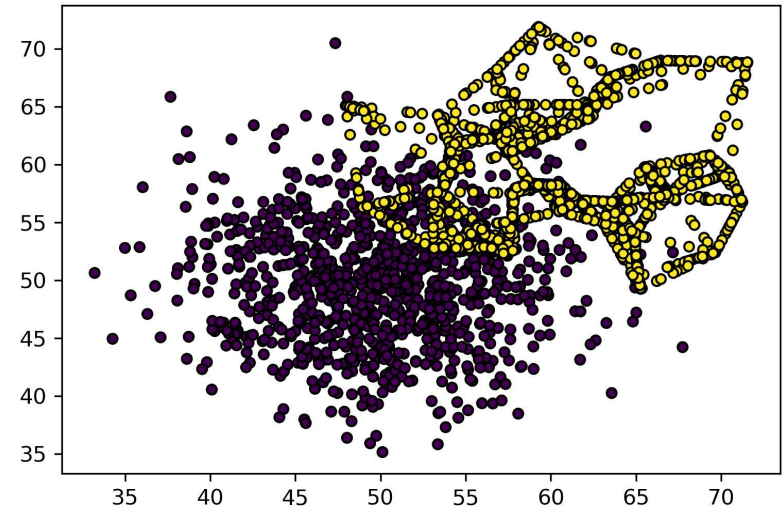
, $k =$

SMOTE

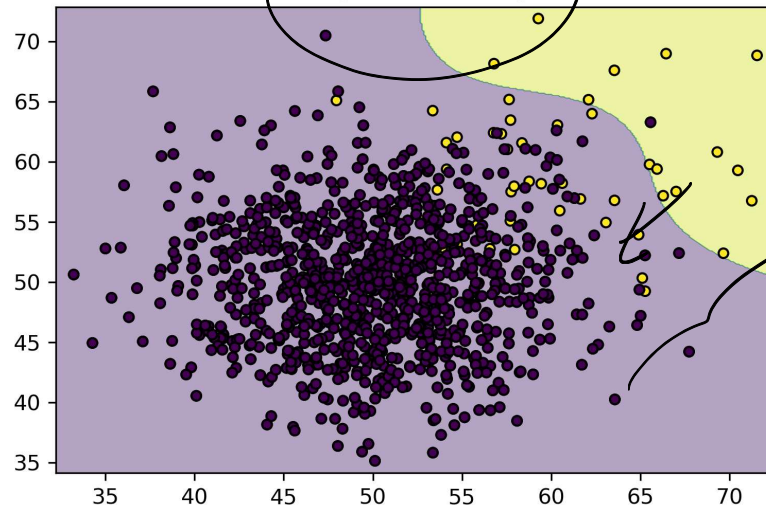
Original Data



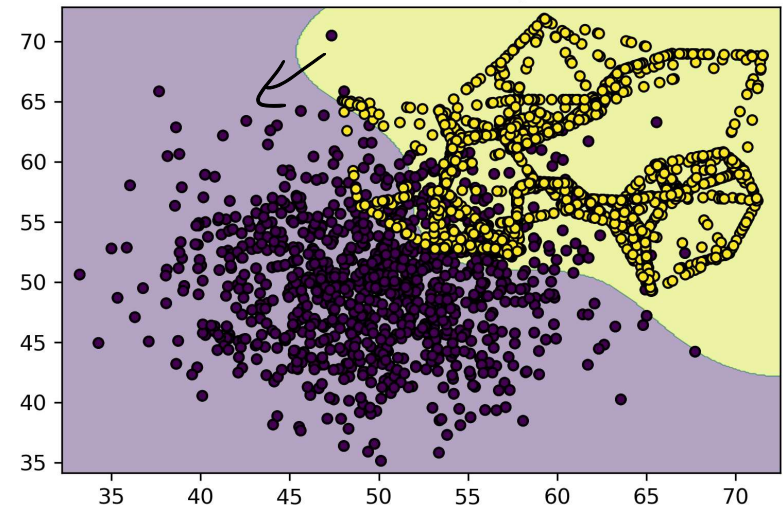
SMOTE Data



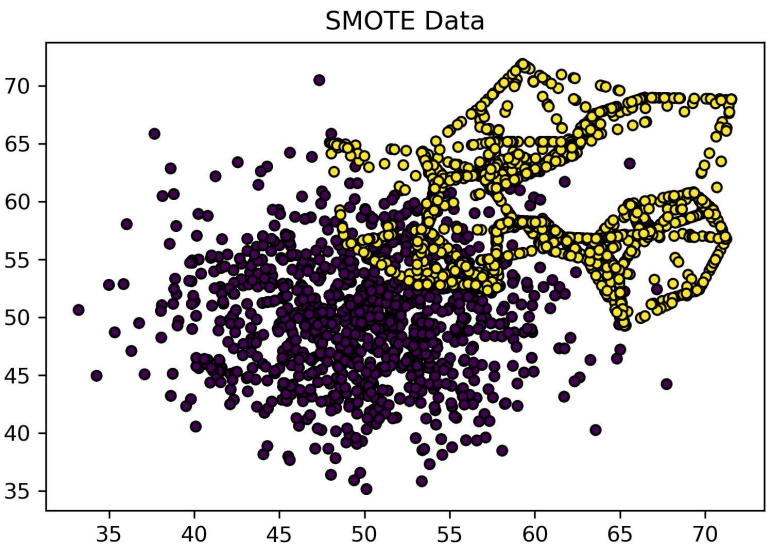
Original Boundary



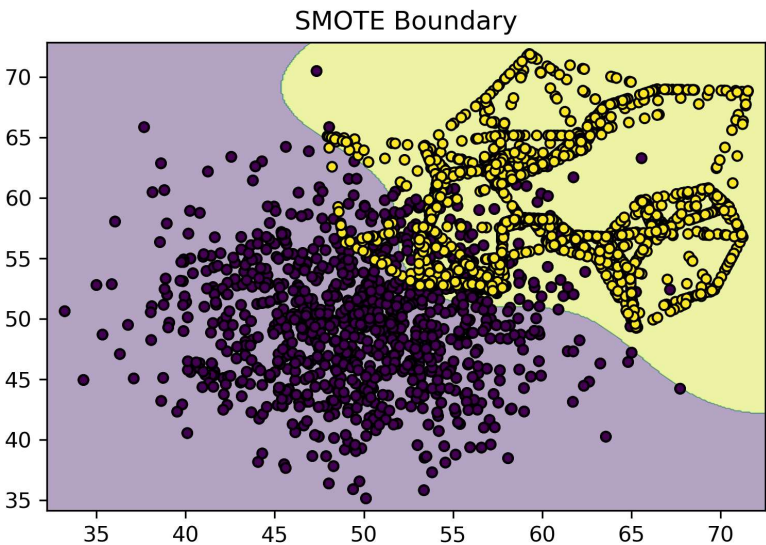
SMOTE Boundary



SMOTE - Accuracy

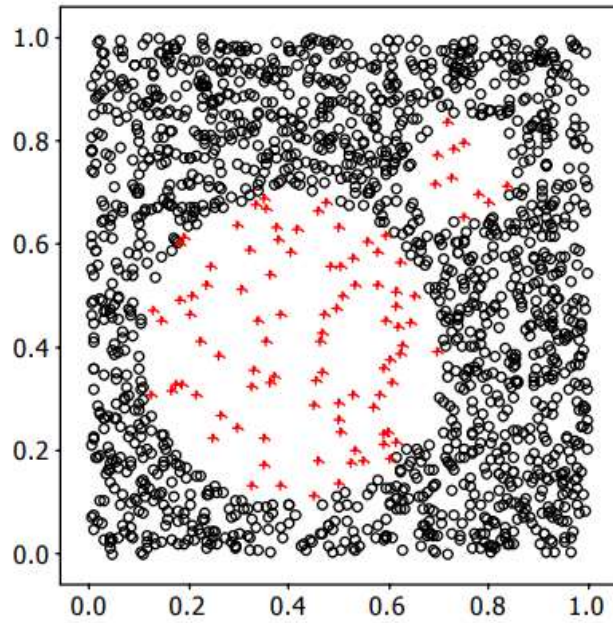


Yellow Accuracy	Purple Accuracy	Accuracy	Balanced Accuracy
0.28	0.999	0.981463	0.6395

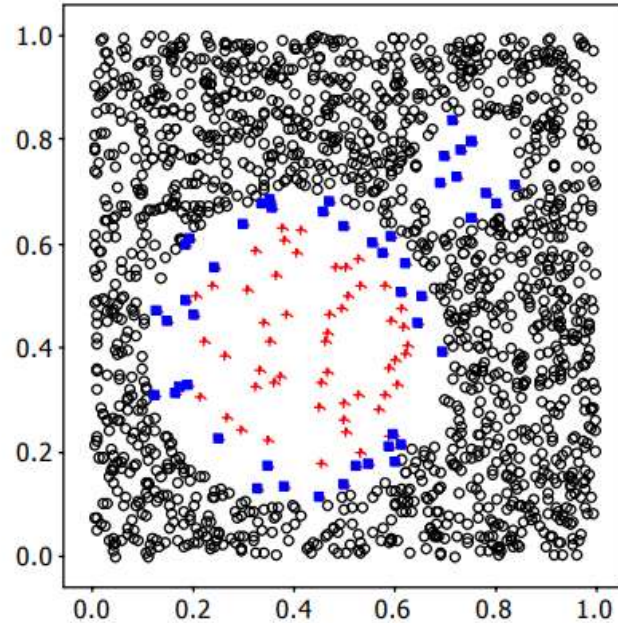


Yellow Accuracy	Purple Accuracy	Accuracy	Balanced Accuracy
0.96	0.864	0.866341	0.912

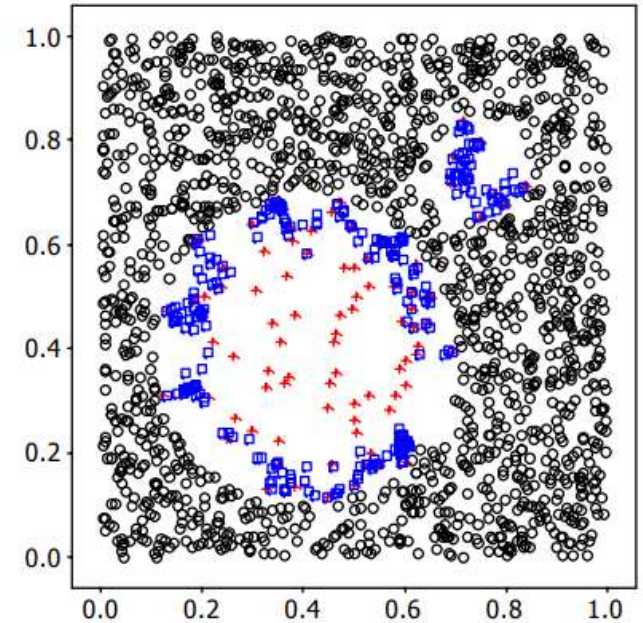
Borderline-SMOTE



The original data



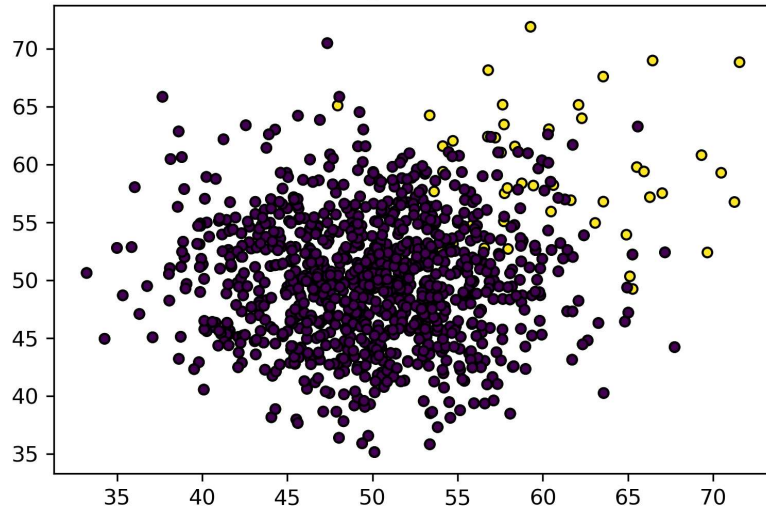
borderline minority



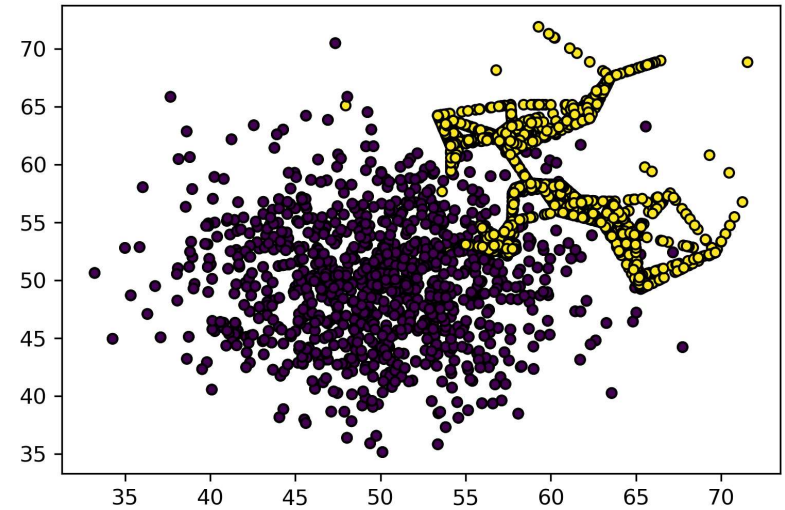
synthetic minority

Borderline-SMOTE

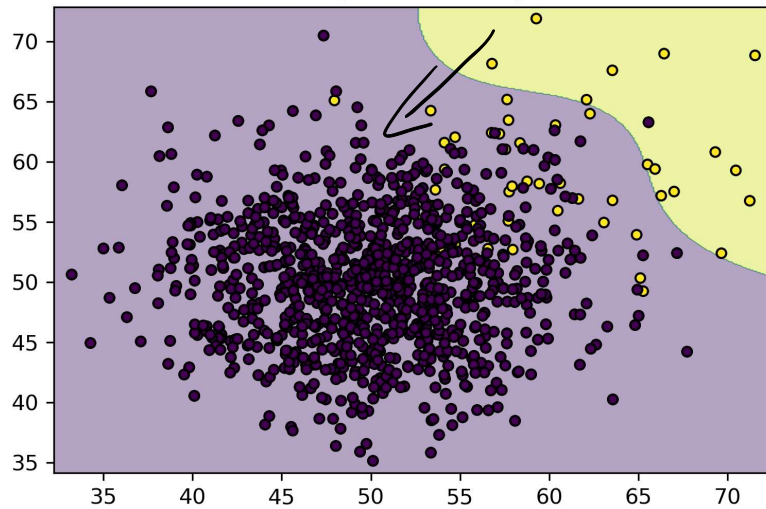
Original Data



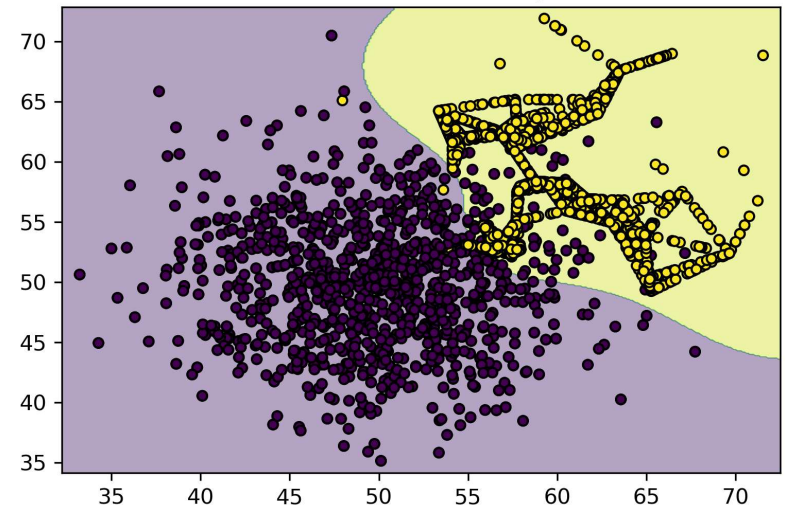
DBSMOTE Data



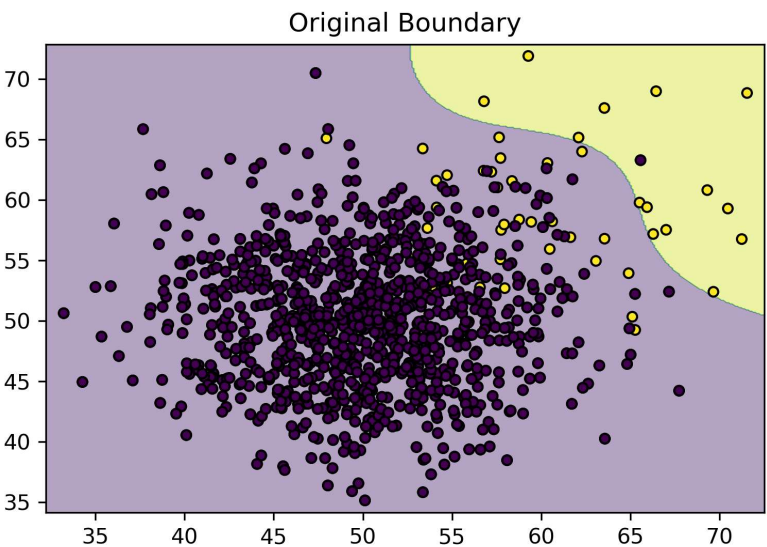
Original Boundary



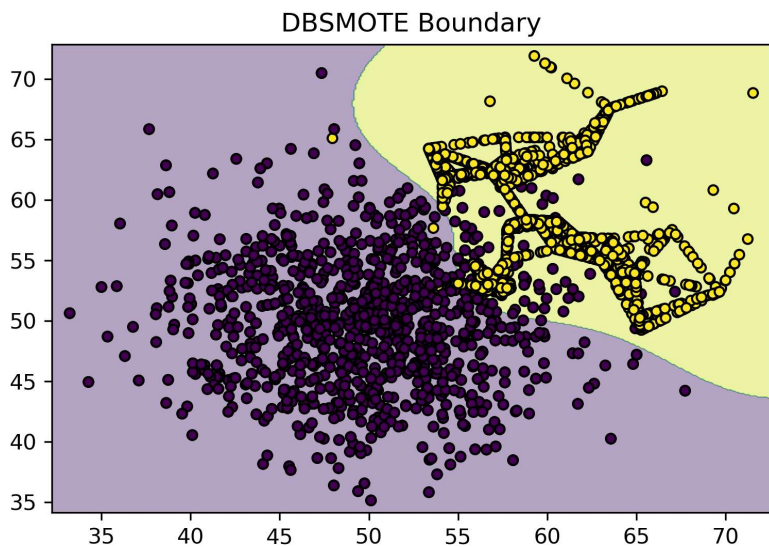
DBSMOTE Boundary



Borderline-SMOTE



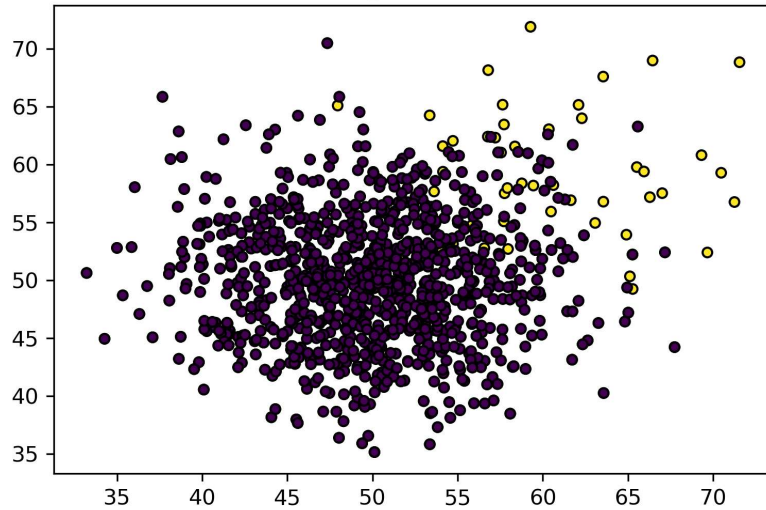
Yellow Accuracy	Purple Accuracy	Accuracy	Balanced Accuracy
0.28	0.999	0.981463	0.6395



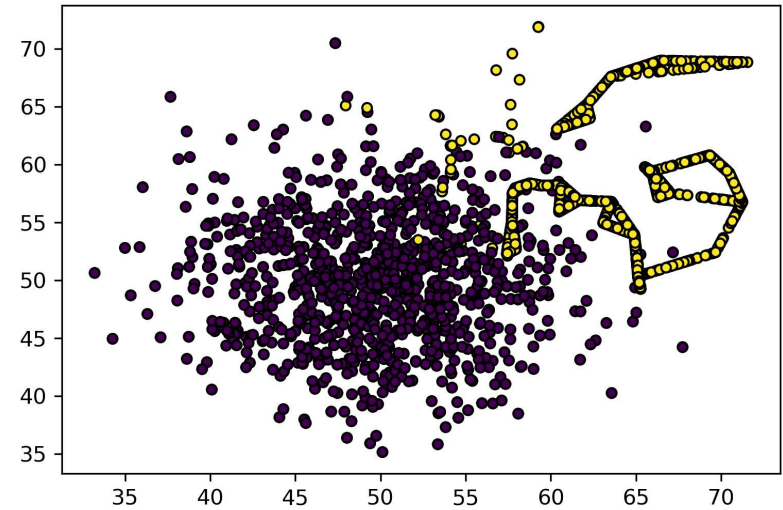
Yellow Accuracy	Purple Accuracy	Accuracy	Balanced Accuracy
0.86	0.919	0.917561	0.8895

Kmean-SMOTE

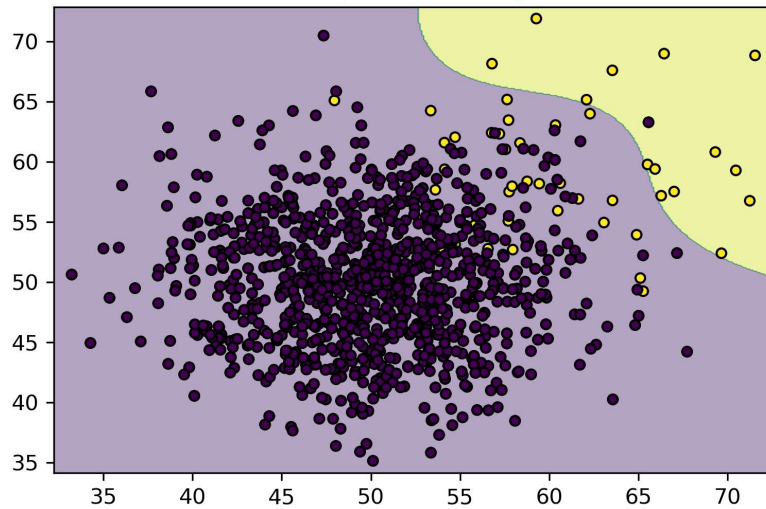
Original Data



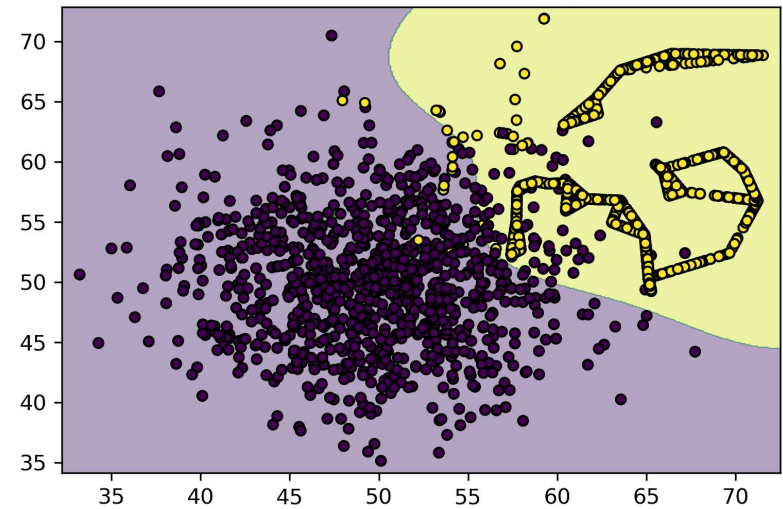
Kmean-SMOTE Data



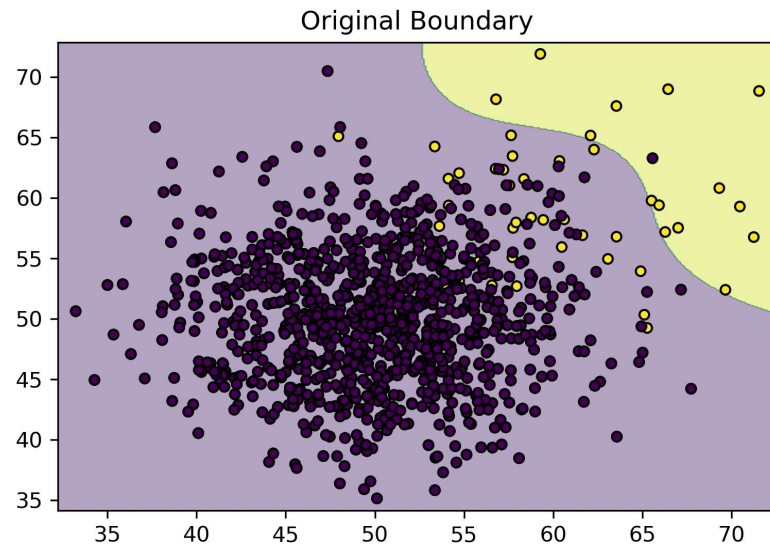
Original Boundary



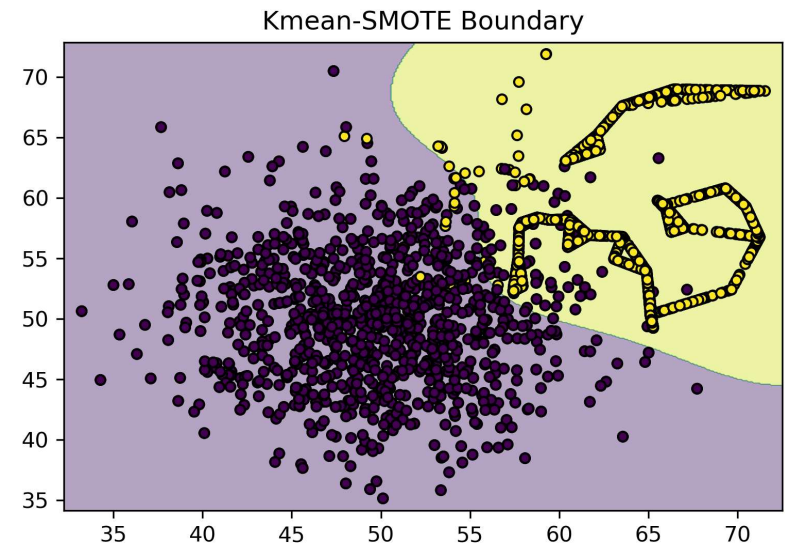
Kmean-SMOTE Boundary



Kmean-SMOTE Accuracy



Yellow Accuracy	Purple Accuracy	Accuracy	Balanced Accuracy
0.28	0.999	0.981463	0.6395



Yellow Accuracy	Purple Accuracy	Accuracy	Balanced Accuracy
0.82	0.94	0.937073	0.88

Research Project: Data Oriented

- Find a dataset
- Propose a method
- Show that it works better than popular methods

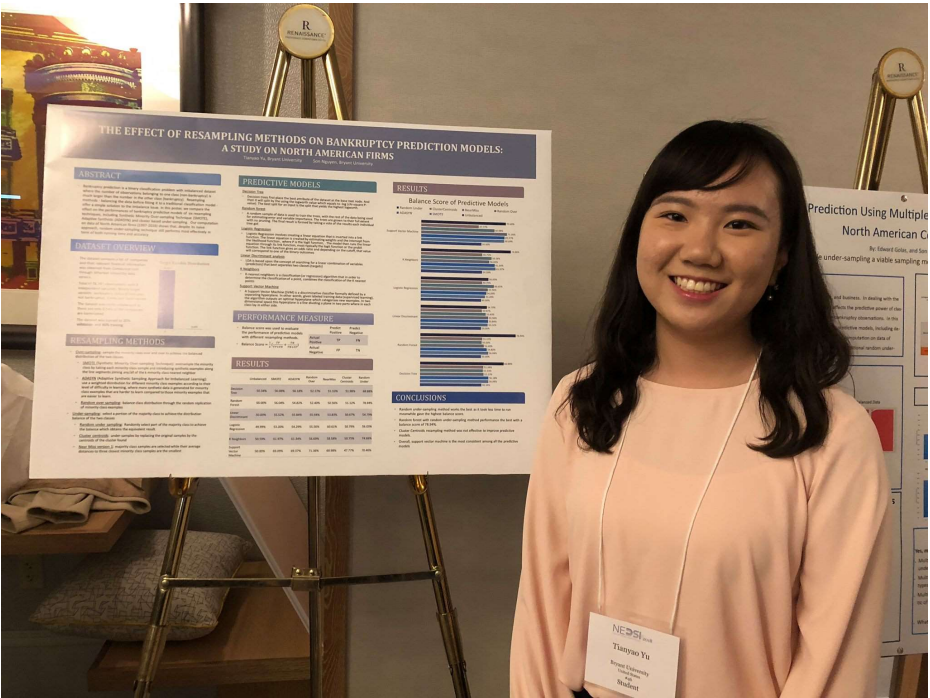
Research Project: Methodology Oriented

- Propose a method
- Find several datasets
- Show that your method works better than popular methods on the datasets

NEDSI 2021

- Northeast Decision Sciences Institute Annual Conference will be held March 26-27,
- Abstract Submission on or before January 16, 2021

NEDSI in the past



NEDSI in the past

